

Preparatory activity and the expansive null-space

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Abstract

The study of the cortical control of movement experienced a conceptual shift over recent decades, as the basic currency of understanding shifted from single-neuron tuning towards population-level factors and their dynamics. This transition was informed by a maturing understanding of recurrent networks, where mechanism is often characterized in terms of population-level factors. By estimating factors from data, experimenters could test network-inspired hypotheses. Central to such hypotheses are ‘output-null’ factors that do not directly drive motor outputs yet are essential to the overall computation. In this Review, we highlight how the hypothesis of output-null factors was motivated by the venerable observation that motor-cortex neurons are active during movement preparation, well before movement begins. We discuss how output-null factors then became similarly central to understanding neural activity during movement. We discuss how this conceptual framework provided key analysis tools, making it possible for experimenters to address long-standing questions regarding motor control. We highlight an intriguing trend: as experimental and theoretical discoveries accumulate, the range of computational roles hypothesized to be subserved by output-null factors continues to expand.

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I dedicate this review to my friend and colleague Krishna Shenoy, who died as we were completing the writing. That act of writing brought our long intellectual relationship full circle. Our final days together were spent much like our initial days: sitting in a room talking about preparatory activity. Pondering the nature of preparatory activity had sent us down the long path of trying to understand the chain of neural events that produces voluntary movement. We saw preparation as the 'first cog' in that causal chain. Preparatory activity must somehow have, wrapped up in it, the information necessary to generate time-varying patterns of muscle activity that will soon occur. Trying to figure out how that could work was the first cog in our personal journey – one whose various branches trace back to the initial question that consumed us: what does preparatory activity do? Our adoption and promotion of other ideas – dynamics, null spaces, factors – were a straightforward consequence of our desire to answer that question. We hope that this review brings readers on a similar journey.

Introduction

The neural basis of movement preparation was first studied using instructed-delay tasks (see refs. 1,2 for early instances of this task design). Such tasks often involve reaching movements, with a variable delay (the preparatory period) separating reach-target onset from a subsequent visual go cue (Fig. 1a). The original intent of this design was that 'preparatory' neural activity, during the delay, could be presumed to subserve a movement-readying computation³. 'Execution-related' neural activity, during movement, could be presumed to provide descending motor commands. Preparatory activity is present in a large percentage of motor-cortex neurons (for example, ref. 4), including those in primary motor cortex (M1) and the adjacent dorsal premotor cortex. Preparatory activity causes no immediate change in muscle activity. Instead, preparatory activity relates to future movement: it reflects movement identity^{1,3–6}, anticipates categorical errors¹, and predicts movement variations⁷ and reaction times^{8,9}. These observations, combined with perturbation experiments^{10,11}, support the hypothesis that preparatory activity is an essential part of the causal chain that produces voluntary movement^{6,12,13}.

As experiments leveraged larger numbers of behavioural conditions (for example, many reach types), it became clear that 'pure'

preparatory responses are rare; nearly all neurons with preparatory activity also exhibit execution-related activity^{6,14} (for example, Fig. 1b). Execution-related activity presumably exists to have eventual downstream impacts (via one or more synapses) on muscle activity. Why then does preparatory activity – in the very same neurons – lack that downstream impact? Put differently, if preparatory activity exists to causally influence movement, what keeps that causal flow at bay during the delay? What then allows preparation to influence execution after the go cue? As experimenters pursued answers to these questions, their efforts became entwined with a conceptual shift that was changing the nature of hypotheses, the language used to express them, and the analyses one could perform. The goal of this Review is to document how the study of preparatory activity both benefited from and informed this conceptual shift.

Traditionally, experimenters attempted to characterize the 'important' properties of single-neuron responses first, providing a foundation for inferring the population response and population-level computation (see refs. 15,16 for examples of this approach in motor cortex). This bottom-up strategy can often succeed; for instance, it has been used to explain behavioural phenomena in terms of sensory-cortex responses (see refs. 17,18 for examples). Yet a pure bottom-up strategy encountered impediments in motor and cognitive areas^{19–21}; the complexity and heterogeneity of single-neuron responses²² made them difficult to characterize. Heterogeneous responses were initially viewed as messy approximations to a simpler ideal. As experimental resolution improved, it seemed increasingly likely that response complexity was intrinsic to the underlying computation. Thus, a new strategy emerged: characterize computation first at the population level^{6,23–27,28}, then determine how single-neuron responses reflect population-level computation.

In motor cortex, single-neuron response complexity is now seen not as a design imperfection but as the outcome of computations that employ a null-space. Suppose each neuron contributes to the eventual output with a given weight, which might be positive, negative or zero (and might reflect monosynaptic or polysynaptic connections). The readout is thus derived from a weighted sum of population activity. Some patterns of population activity occur within an output-potent space; that is, they change the value of the

Fig. 1 | Preparatory and execution activity in empirical data and an artificial spiking network. **a**, Basic structure of instructed-delay tasks, illustrated using the 'maze' task, in which reaches to a target must avoid barriers. Each trial begins when the monkey touches and holds a central spot. After a few hundred milliseconds, barriers and a reach-target appear, providing the instruction. The reach must be withheld until a go cue, which occurs after a variable delay (averaging ~600 ms). The reach-target jitters slightly in place when it first appears; cessation of jitter provides the go cue. The 'preparatory period' is the delay from target onset to go cue. Reaches typically begin soon (~300 ms) after the go cue. A 'condition' is a set of stimuli (for example, barriers and target) evoking a specific reach type (for example, a curved reach to an upward target). A trial is one repetition of that condition. Single-neuron firing rates are typically computed by averaging across trials within a condition. **b**, Example responses of five motor-cortex neurons recorded during the maze task, from monkey²⁴. Each trace plots the trial-averaged firing rate for one condition. Preparatory-period neural activity arises shortly after target onset (first arrow). Traces are coloured (separately for each neuron) on the basis of the intensity of preparatory activity within each condition. Execution-related activity begins after the go cue (not shown), ~150 ms before reach onset (second arrow). **c**, Illustration of how factors (blue circles) relate to connectivity in an artificial spiking network⁷⁷. The activity of all neurons is

described by the vector $\mathbf{r}(t)$. An entry of $\mathbf{r}(t)$ is one if that neuron recently spiked and zero otherwise. Each factor is a weighted sum of those spikes: $x_i(t) = \mathbf{w}_i^T \mathbf{r}(t)$, where neural dimension $\mathbf{w}_i$ contains weights for the i^{th} factor. Unlike spiking neurons, factors are continuously valued and their evolution is readily described using a state space (inset). Given recurrent connectivity, factors both are determined by neural activity (straight grey arrows) and influence it at the next moment (curved grey arrows). The impact of a factor on neural activity is $\mathbf{u}_i x_i(t)$, where $\mathbf{u}_i$ is a 'factor-tuning' dimension. Dimensions $\mathbf{w}_i$ and $\mathbf{u}_i$ reflect synaptic connectivity, determined during network training. Network outputs (the 'readouts') are weighted sums of neural activity, $z_j(t) = \mathbf{b}_j^T \mathbf{r}(t)$, where $\mathbf{b}_j$ is the readout dimension for the j^{th} readout. **d**, Spikes from one neuron in such a network. This network was trained to produce, as readouts, the empirical activity of two muscles (biceps and deltoid) during reaching. Spikes are shown for multiple trials (one per row) across different conditions (reaches in eight directions). Colour-coding as in panel **b**. **e**, Trial-averaged firing rate of the simulated neuron from panel **d** (top), and one of the two network readouts (bottom). This readout was trained to mirror empirical biceps activity; each trace shows that readout on a single trial for one condition. Simulated-neuron firing rates are colour-coded as in panel **b**, and readouts are given the same colour-coding. Panel **b** adapted from²⁴, Springer Nature Ltd. Panels **d** and **e** adapted with permission from⁷⁷, Elsevier.

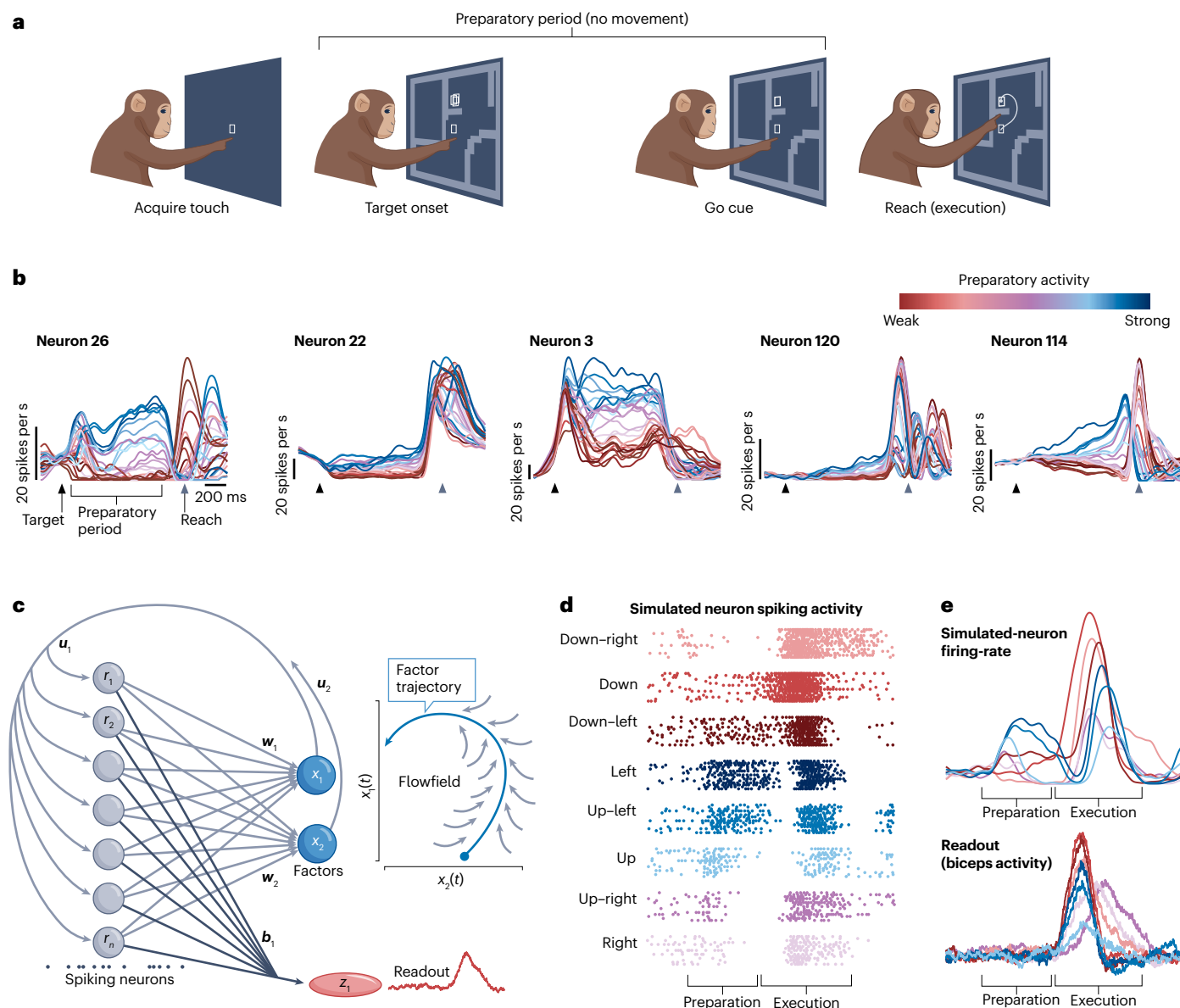
weighted sum. Other patterns of population activity occur within a null-space; that is, the weighted sum remains unchanged and the readout remains constant^{29–31}. A key realization was that, although null-space activity might initially seem pointless, it can be central to network computation^{23,24}. This realization reflected a conceptual shift that occurred near-simultaneously across multiple groups studying a variety of areas, aided by an improving understanding of how recurrent networks compute.

In this Review, we focus on the role played by the study of motor cortex within this broader conceptual shift. Motor-cortex neurons are as little as one synapse from spinal motor neurons³², and individual-neuron spikes can measurably impact muscle activity^{33,34}. Muscle activity is readily generated by microstimulation^{35–37} almost anywhere in motor cortex. These observations highlighted the need to explain how preparatory activity can exist without causing muscle activity. Experimenters were also wrestling with another conundrum: during

execution, motor-cortex activity patterns look quite different from muscle-activity patterns. We review how the null-space concept provided potential explanations for both conundrums, and how seemingly odd features of individual-neuron responses began to make sense in the context of a null-space. We also review how thinking in terms of neural subspaces – both output-null and output-potent – allowed experiments to address long-standing questions and facilitated predictions that were tested using modern perturbation approaches. Looking forward, we discuss how the null-space is likely to play multiple computational roles, and how the ability to sustain large movement repertoires may depend upon an expansive null-space.

Traditional questions through a network-centric lens

A striking property of single-neuron responses in motor cortex is their complexity and heterogeneity (Fig. 1b). Preparatory activity exhibits



one pattern of tuning, whereas execution-related activity exhibits another (Fig. 1b, Neurons 26 and 114). During execution, multiple response peaks are often present, with amplitudes and phases that vary with reach condition (Fig. 1b, Neurons 26, 120 and 114). A neuron's 'preferred condition' (the reach type evoking the largest response) is thus typically inconstant, differing between preparation and execution and evolving throughout execution. Each neuron's response tends to be complex in its own way. Response complexity and heterogeneity are prominent across multiple tasks including reaching^{20,22}, cycling a hand-held pedal³⁸ and force-tracking³⁹.

The full extent of response complexity was initially underappreciated. Given the crystalline beauty of receptive-field organization in visual areas, motor-cortex neurons were expected to be similarly well-behaved. Network models reflected this expectation: they tended to be hand-constructed with sensibly organized single-neuron tuning. For example, preparatory activity might form a hill within a field of spatially tuned neurons, each 'voting' for a different preferred movement^{40–42}. A rising hill would trigger the specified movement, much as is proposed to occur when the superior colliculus generates a saccade^{43,44}. The challenge was thought to be determining which parameters were represented in the field and selected by the hill. Models of this type do not need a null-space; threshold-style non-linearities keep preparatory activity from flowing downstream.

Yet it steadily became clear that neural activity cannot be summarized via preferences for external parameters^{19–22,45–47}. Furthermore, although neural activity exhibits many muscle-like features^{48–54}, it does not reliably mirror muscle activity^{22,23,38,45}. These realizations prompted interest in a different class of network solutions: ones that emerged when initially unstructured recurrent networks were trained to generate motor commands. Early motor-cortex models of this type were small^{22,55} or employed limited dynamics⁵⁶, yet were encouraging because they displayed realistically 'messy' single-neuron responses. Concurrently, theorists were exploring general properties of randomly connected networks^{57–59}. Even when untrained, such networks exhibited intriguingly realistic properties, including variable spiking⁵⁹, input-driven quenching of variability^{60,61}, dynamics capable of generating complex patterns^{57,62,63}, and heterogeneous single-neuron responses⁶⁴. These observations motivated a desire to understand the computational strategies – including null-spaces – available to such networks^{65,66}. As better tools were developed for training recurrent networks^{64,67}, it became easier to compare network solutions with data. Data analysis started to borrow from, and anticipate, methods for reverse-engineering network solutions. This created a fundamental shift in our view of a single-neuron's response: away from the idea of voting for a preferred parameter, and towards the idea that each neuron's response is an incomplete reflection of population-level factors.

The factor-dynamics framework

The resulting conceptual framework has been termed the "dynamical systems view"^{66,68,69} or "computation through neural population dynamics"¹³. In this Review, we use 'factor-dynamics framework' to highlight that, although network dynamics govern individual-neuron spiking^{70,71}, they are most succinctly described in terms of population-level factors. When theorists reverse-engineer a network with recurrent connectivity, they compute weighted sums of population activity (for example, refs. 25,72,73). A set of weights, with one weight per neuron, constitutes a neural dimension. The resulting weighted sum of neural responses is referred to as 'activity in that dimension'. It is often possible to intelligently choose a set of neural dimensions with the following property:

activity in those dimensions provides a full account of how the network does its job. We use the term factors to refer to this useful set of weighted sums.

Neural dimensions, and the factors they define, have historically been chosen in a variety of ways. Some theoretical studies identify factors on the basis of network connectivity (for example, ref. 72). Others analyse network responses (for example, ref. 24). Some empirical studies estimate factors by applying dimensionality reduction to trial-averaged firing rates (for example, ref. 23). Others infer factors by treating them as latent variables that probabilistically influence single-trial spiking (for example, refs. 74,75). Although it is useful to have a range of approaches for factor estimation, this can also seed uncertainty regarding what factors truly 'are'. It is important to dispel such uncertainty, because a mechanistically grounded definition of factors is necessary to understand how there can exist output-potent factors that directly and causally impact readouts, and output-null factors that do not. To provide that definition, this Review leverages a version of the factor-dynamics framework that applies to a large class of recurrent spiking networks, where factors are defined as linearly weighted sums of neural activity. This definition of factors reflects the basic assumption that synaptic inputs are summed (for example, two excitatory post-synaptic potentials roughly add) before determining whether a spike occurs. Despite this linear definition of factors, networks of this type can perform a vast array of highly non-linear computations, and neural activity can occupy very non-linear manifolds. In explaining how computation is performed, the factor-dynamics framework embraces three tenets.

First, the activity of a population of seemingly noisy spiking neurons can be summarized by continuously valued factors (Fig. 1c, blue circles). Factors capture the reliable signals – all signals that are not just spiking noise – flowing through the network's connectivity. This summary works because each neuron's synaptic inputs derive from a weighted sum of the spikes of all other neurons. Of all the weighted sums that could exist, only a subset carries reliable, meaningful, signals. Each factor is one such weighted sum, and a linear combination of factors describes the meaningful component of each neuron's input. Factors may be numerous^{22,39,69,76}, but if we have N neurons and K factors then $K \ll N$. Factors help to characterize how the spiking of N neurons, at one moment in time, causes their activity at the next moment. Rather than considering all synapses and all N^2 spike-based interactions, one can consider K factors ($K = 2$ in Fig. 1c). One set of neural dimensions (w_i) describes how factors depend upon spiking activity (straight arrows in Fig. 1c). Another set of neural dimensions (u_i) describes factor tuning: how each neuron's inputs (and thus activity) reflect a weighted sum of factors⁷⁷ (curved arrows in Fig. 1c). If one knows the current pattern of spikes, one can compute the present value of the factors using w_i . On the basis of factor tuning (u_i), one can then compute the probability, at the next moment in time, that each neuron will spike⁷⁷. Although spiking is typically noisy, factors are not. Each factor is a weighted sum across many neurons; unreliable aspects of spiking largely average out⁷⁷.

The second tenet is that characterizing factor-level dynamics is necessary and often sufficient to understand computational mechanism. By dynamics, we mean how the factor-values at the present moment determine the factor-values at the next moment, through the recurrent loop from factors, to neurons, then back again (Fig. 1c). Factor-level dynamics are often summarized via a flow-field in state space (Fig. 1c, inset). The present value of the factors is referred to as the neural state. As this state evolves, it traces a neural trajectory, sculpted by the dynamics. Employing the factor level is necessary

because tracing dynamics spike by spike and neuron by neuron is infeasible. The factor level can be sufficient because factors, if properly defined, constitute all reliable signals flowing through the recurrent connectivity. Owing to this unifying factor-level description, one can construct equivalent models built from spiking neurons (see ref. 77), rate-based neurons (see ref. 24) or abstracted state variables (see ref. 23). Such models can perform the same computation, in the same way, using the same factor-level dynamics.

The final tenet is that many factors are output-null; their value has no direct impact on the readout the experimenter is observing. Factors constitute all reliable, meaningful, weighted sums of spikes. The set of such weighted sums used by the network to perform its computation is typically larger than the set used by the readouts. For example, responses in Fig. 1d,e were generated by a network that uses dozens of factors to generate the activity of just two muscles⁷⁷. Most neurons drive the muscles, via weights described by readout dimensions (Fig. 1c; arrows labelled $\mathbf{b}_i$). Most neurons also show changes in their spike-rate during preparation. Yet muscle readouts remain constant because ‘preparatory’ factors are output-null. Importantly, output-null factors are not ‘computationally null’: they determine the future evolution of output-potent factors. This idea – factors that are output-null yet mechanistically critical – was central to hypotheses that viewed factor-dynamics as an alternative to parameter-encoding^{23,24}.

Since its inception, the factor-dynamics framework has been associated with the use of dimensionality reduction methods^{74,78–80}, such as principal component analysis (PCA). Such methods estimate factors from data, allowing tests of model-derived hypotheses (see refs. 23,81 for examples). Ideally, empirical analysis would leverage knowledge of connectivity, making it possible to estimate both $\mathbf{w}_i$ (how factors depend on activity) and $\mathbf{u}_i$ (how activity depends on factors). In many artificial networks, $\mathbf{w}_i$ and $\mathbf{u}_i$ can be directly computed from synaptic connectivity. Importantly, they are not identical (see refs. 72,73,82 for situations where this matters). The set of $\mathbf{u}_i$ describes the activity subspace: what an experimenter would discover when applying dimensionality reduction to firing rates. The set of $\mathbf{w}_i$ describes the dynamics subspace⁸²: neural dimensions where perturbations of activity will impact future computation. Without knowledge of connectivity, $\mathbf{w}_i$ cannot be estimated. Yet factors still can be. Dimensionality reduction estimates the activity subspace spanned by $\mathbf{u}_i$. Because each neuron’s activity depends on the factors (via the weights in $\mathbf{u}_i$) one can invert this relationship to infer what the factors must have been given the observed neural responses. This is exactly what most dimensionality reduction methods seek to do.

Shifting from neuron-level thinking to factor-level thinking

The factor-dynamics framework helped to demystify seemingly odd single-neuron response properties. In factor-based models, a neuron’s response is typically determined by multiple factors, some output-potent and many output-null. Such ‘mixed’ individual-neuron responses can be confusing on their own, yet make sense once the underlying factors are characterized. In turn, characterization of factor-level dynamics provided a way to understand solutions embedded in large non-linear networks. Even in spiking networks, whose elements are strongly non-linear, it is possible to define linear relationships between neural activity and factors⁷⁷. The impact of non-linearities is captured at the level of dynamics, and computation can often be understood in terms of those dynamics.

Our field’s shift towards factor-dynamics ideas can seem sudden; hypotheses and analyses differ sharply between present

studies of motor control and those from two or three decades ago⁸³. Yet dynamics-centric ideas date to the earliest studies of movement^{84,85}. Many oculomotor models focused on controller dynamics^{86–88}, without insisting that neurons make sense individually²⁶. Kwan et al. proposed control via recurrent-network dynamics⁵⁵, and Fetz used such networks to embrace single-neuron response complexity^{20,56}. Motor control via dynamics, without explicit movement-parameter representations, was proposed on philosophical grounds^{89,90}. Kalaska and colleagues stressed that motor-cortex activity reflects ‘generative details of movement’ rather than high-level parameters^{48,91}, and Todorov, Scott and Cisek solidified the tenet that single-neuron tuning should be understood in the context of movement-generating network strategies^{42,49,51,92–94}. Furthermore, it was appreciated that motor cortex is a point of convergence for multiple forms of recurrent connectivity, including internal and external feedback loops^{95–98}, making it a natural location to observe dynamics. Null-spaces were critical in linking the idea of control via dynamics to network-level solutions that could explain neural responses. Output-null factors are now hypothesized to subserve a variety of important computational roles (Fig. 2), as will be described in subsequent sections. One specific role – movement preparation – was central to establishing the null-space concept.

Early conceptions of preparatory activity

When motor-cortex preparatory activity was first documented, its purpose was suspected to be subthreshold modulation of spinal motor neurons and spinal reflexes¹. In this view, preparatory activity directly impacts outputs, but that impact is invisible when motor neurons are silent. An alternative conception soon arose: preparatory activity might reflect internal ‘programming’ that readies movement^{3,99,100}, with no immediate change in motor output. In support of this, even when muscles are active (for example, when holding the hand on the screen in Fig. 1a) they still do not typically change their activity during preparation^{2,9,14}. Spinal reflexes are also largely unchanged during preparation¹⁰¹. These observations supported an idea that later became embedded in the factor-dynamics framework: the primary purpose of preparatory activity is to causally impact future execution^{4,8,9,14,42}. If so, what defers this impact until after the go cue?

One possibility was sequestration of preparatory and execution-related activity into separate motor-cortex populations – that is, preparatory-only neurons and execution-only neurons. Certainly there do exist execution-only neurons: preparatory activity becomes weaker in sulcal M1 and spinal interneurons¹⁰² and is largely absent in spinal motor neurons (hence the lack of muscle activity). Yet this only highlights the conundrum faced when considering the large population of motor-cortex neurons that are active during both preparation and execution. These ‘preparation-and-execution’ neurons are connected, monosynaptically and polysynaptically, with downstream execution-only neurons in sulcal M1 and the spinal cord. What mechanism allows their execution-related activity, but not their preparatory activity, to flow downstream?

One hypothesis was that preparatory activity might be a subthreshold version of execution-related activity. This possibility was incorporated into ‘rising hill’-style models^{40–42}, discussed above. Yet it became increasingly clear that preparatory activity was not a weaker form of execution activity^{6,103}; as discussed above, a neuron’s ‘preferred condition’ is not preserved between preparation and execution. An alternative suggestion was that strong intracortical inhibition might suppress output neurons until execution. This hypothesis predicts inhibitory neurons that are strongly active during the preparatory period, then become

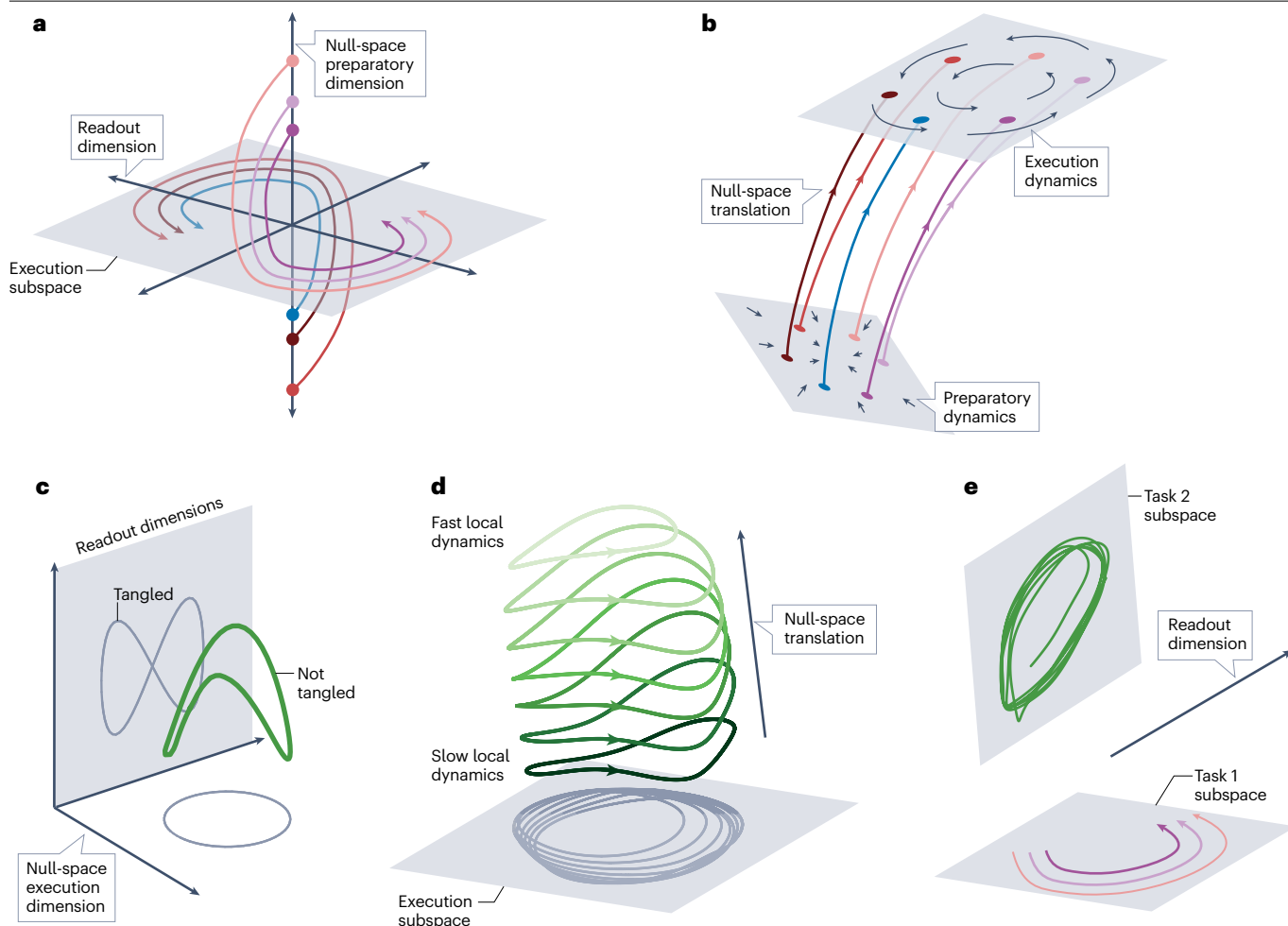


Fig. 2 | Output-null factors are hypothesized to serve multiple purposes.

a, A mechanism by which preparatory activity could avoid impacting present muscle activity, yet still specify upcoming movement. In this illustration, preparatory activity develops within a single output-null dimension and each coloured trace plots the trajectory for one reach condition, starting from the final preparatory-period neural state. Just before movement onset, trajectories ‘rotate’ into an execution subspace. The execution subspace overlaps with a muscle-readout dimension. Because the spread of preparatory neural states (circles) is orthogonal to the readout dimension, preparatory-period activity does not prematurely impact muscle activity. Yet via trajectory rotations, the preparatory-period state impacts what will occur in the execution subspace. **b**, A large output-null translation in state space is hypothesized to trigger the transition from preparatory dynamics to execution dynamics, and thus to initiate a reaching movement. In large networks, factor-level dynamics (black flow-fields) can be strongly non-linear. Different regions of state space can thus ‘contain’ different local dynamics, making it possible to transition from preparatory dynamics to execution dynamics using a translation. The translation occurs in a dimension orthogonal to those in panel **a**, and is what creates the transition from relatively static preparatory states to rotating execution trajectories.

c, Output-null factors are hypothesized to reduce trajectory tangling. A tangled trajectory is one that crosses itself (or comes close). Without a null-space, neural trajectories would occupy only the readout dimensions (grey plane) and would thus resemble muscle trajectories, which may be tangled. The output-null factor allows the neural trajectory, in three dimensions, to avoid crossing itself. A consequence of the presence of the output-null factor is that the trajectory becomes more circular. Indeed, it is exactly a circle in the two dimensions that provide the ‘floor’ of this plot. These simplified neural and muscle trajectories resemble those seen in the cycling task, where monkeys continuously turn a hand-held pedal. **d**, A translation in an output-null dimension is hypothesized to select amongst variations of a given movement. In this example, lower trajectories experience slower local dynamics, and produce slower cycling. Higher trajectories experience faster local dynamics, and produce faster cycling. **e**, One network could perform two unrelated tasks – perhaps cycling and reaching – using non-overlapping sets of factors. Neural trajectories for the two tasks (coloured traces for task 1 and green trace for task 2) would thus occupy orthogonal subspaces (grey planes). Trajectories for both tasks can still impact the same readout dimension. This ‘flexibility-via-subspace’ strategy requires dimensions in addition to the readout, and thus a null-space.

quiet during execution. This would mirror a similar phenomenon that is well-established in the oculomotor system¹⁰⁴. Yet the predicted response pattern is rare in motor cortex, and is least common – not most common – for narrow-waveform neurons (a population enriched with

inhibitory interneurons)¹⁰³. Blanket inhibition of motor-cortex output would also probably be problematic from a design perspective: it would not be possible to prepare while maintaining a posture, or to prepare one movement while making another.

Null-spaces provide a potential mechanism

Before it became possible to reliably train recurrent networks on motor tasks, it was known that they could generate rich activity patterns that might form building blocks of outgoing motor commands^{57,62,63,105}. It was also appreciated that many of these activity patterns, although essential to network function, would fall in the null-space of network readouts. This was hardly a radical observation – it was typical of how networks worked – suggesting that biological neural networks might do so as well.

A common way of illustrating the null-space idea is to plot the set of preparatory states (one per reach condition) in a state space (Fig. 2a, coloured circles). If the spread of preparatory states is orthogonal to the readout-defining neural dimension, all conditions produce the same readout despite having different preparatory states. This visualization is useful, but does not directly convey the mechanistic reason why null-space activity avoids impacting outputs. That reason is often illustrated by invoking two neurons driving the same downstream target. If neural responses change in opposition, net drive remains constant^{30,106}. This mechanism might seem fragile when considering two neurons, yet for large populations most changes in activity fall in the null-space.

To illustrate why a null-space is natural network property, this Review considers a simple toy model (Fig. 3a) that uses six factors (not shown) to perform a delayed-reach task. The model embodies three assumptions that have become common in network models of reaching. First, preparatory activity initializes dynamics, determining what will occur during execution. Second, factor-level dynamics contain a rotational component, creating an order-preserving flow of activity from some neural dimensions into others (as in Fig. 2a, coloured lines). Last, individual-neuron responses are incomplete reflections of factor-level dynamics. These features were first implemented in idealized models embodying an educated guess regarding recurrent-network solutions^{23,62,107}. There now exist multiple network-style models that share most or all of these features^{24,81,108–110}.

In the present simulation, two ‘preparatory factors’ are active during preparation and become zero during execution. Four ‘execution factors’ are zero during preparation and become active during execution. Factors display simplified rotational dynamics, with phase and amplitude determined by the preparatory factors (as in Fig. 2a). We do not model the source of dynamics (presumably a combination of cortical recurrence, multi-area loops and sensory feedback) but simply assume them and simulate firing rates for a 5,000-neuron population. Firing rates are shown for 25 neurons (Fig. 3a, top) for three different reaches in succession. Each neuron’s rate reflects a combination of the six factors. The impact of the i^{th} factor on each neuron is described by neural dimension $\mathbf{u}_i$ (as in Fig. 1c) containing one factor-tuning weight per neuron. We simulated two network outputs (Fig. 3a, red traces). One output corresponds to a muscle that remains inactive, during both preparation and execution, for all conditions in this task. The other output corresponds to a muscle that becomes active during execution. The readouts driving the muscles are weighted sums of single-neuron activity. Readout weights are defined by neural dimensions $\mathbf{b}_{\text{inactive}}$ and $\mathbf{b}_{\text{active}}$. For example, the dimension $\mathbf{b}_{\text{active}}$ contains one weight per neuron, describing that neuron’s net impact (via monosynaptic and polysynaptic connections) on the active muscle. Neural activity is summarized by a vector of firing rates, $\mathbf{r}(t)$. For simplicity, we use vector notation to indicate a weighted sum. For instance, the dot product $\mathbf{b}_{\text{active}}^T \mathbf{r}(t)$ is the weighted sum obtained by multiplying each neuron’s activity by its weight, then summing to obtain the net impact of all neurons on that muscle.

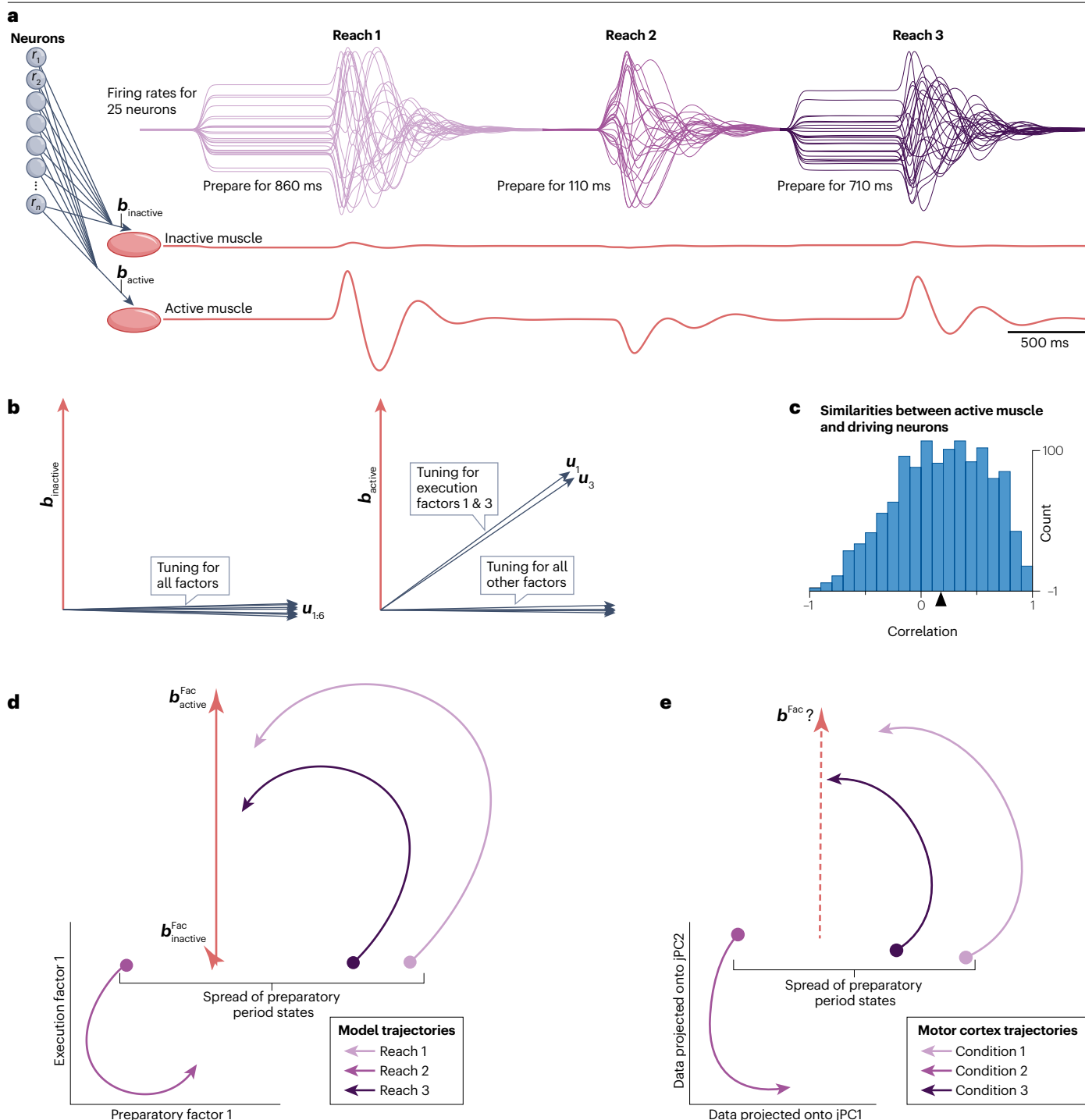
First consider the inactive muscle. We chose $\mathbf{b}_{\text{inactive}}$ randomly to illustrate a counterintuitive point: not only is there no paradox in a muscle remaining silent when upstream neurons are active, this tends to be true by default. The impact of the population response on the muscle is the weighted sum $\mathbf{b}_{\text{inactive}}^T \mathbf{r}(t)$. The impact of the i^{th} factor on the population response is $\mathbf{u}_i x_i(t)$. Thus, the impact of the factor on the muscle is $(\mathbf{b}_{\text{inactive}}^T \mathbf{u}_i) x_i(t)$. Without a learned relationship, the entries of $\mathbf{b}_{\text{inactive}}$ and $\mathbf{u}_i$ are uncorrelated and $\mathbf{b}_{\text{inactive}}^T \mathbf{u}_i \approx 0$ (Fig. 3b, left). Consequently, despite each factor influencing most neural responses, the impact on the inactive muscle averages to near-zero. This property does not require linearity. If outputs are non-linear functions of weighted sums, they remain constant if the weighted sums remain constant. Many other non-linearities can be approximated using a state-dependent $\mathbf{b}$. Unless state-dependence causes $\mathbf{b}$ and $\mathbf{u}_i$ to correlate, the factor remains output-null.

A factor impacts muscle activity only if there exists a relationship between its factor-tuning dimension and the readout dimension, such that $\mathbf{b}^T \mathbf{u}_i \neq 0$. We thus constructed $\mathbf{u}_1$ and $\mathbf{u}_3$ (tuning for two execution factors) such that their entries correlate moderately with $\mathbf{b}_{\text{active}}$ (Fig. 3b, right). Factors 1 and 3 now have a net influence, and the active muscle’s activity is their sum. Importantly, the muscle is driven by diverse single-neuron responses – most with preparatory activity and few resembling muscle activity (Fig. 3c). Yet the influence of all factors, other than $x_1(t)$ and $x_3(t)$, averages out. Preparatory activity is simply a special case of this phenomenon – no dedicated ‘preparatory suppressing mechanism’ is needed.

Testing the hypothesis that preparatory activity occupies a null-space

The null-space hypothesis accorded with some otherwise confusing facts. Under the null-space hypothesis, and the factor-dynamics hypothesis more generally, neurons do not typically form distinct categories of ‘preparatory neurons’ versus ‘execution neurons’, or ‘rightward-reach neurons’ versus ‘leftward-reach neurons’. Each neuron’s response reflects an essentially random mixture of preparatory factors and execution factors. A neuron may thus appear to participate in preparation for some movements and execution for others. The null-space hypothesis also predicts ‘mismatched tuning’: preferred conditions that differ between preparation and execution, as illustrated by the example simulated neurons (Figs. 1e and 4b–f). This mismatch is mechanistically important; without it, readout weights that create muscle activity during execution would inevitably do so during preparation. Single-neuron responses in motor cortex show these predicted properties^{6,14,103} (for instance, Figs. 1b and 4a). The null-space hypothesis further predicts that, despite seemingly haphazard single-neuron tuning, muscle activity should be directly decodable from neural activity. Indeed, it typically can be^{22,38,111–113}.

These observations were suggestive, but a more direct test was sought. The null-space proposition is simple: muscle-activity commands are derived from weighted sums of single-neuron activity, and those weighted sums remain constant during preparation. The most direct test would be to estimate the relevant weights, $\mathbf{b}$, apply them to recorded neural responses, $\mathbf{r}(t)$, and ask whether the resulting weighted sum has the predicted properties. Unfortunately, estimating the weights is challenging. It is possible to identify strong monosynaptic connections (and even disynaptic inhibitory connections), between cortical neurons and spinal motor neurons, using spike-triggered averaging^{33,34}. This approach depends upon single cortical spikes immediately influencing motor neuron spikes, and is thus well-suited to identifying



strong, fast connections. It is unlikely to be well-suited (especially given the technical challenges involved) to estimating each recorded neuron's net drive (including polysynaptic connections) to each muscle.

A more tractable test becomes available if we consider that, under the null-space hypothesis, muscle activity is a weighted sum of factors. This is true because muscle activity is a weighted sum of neural activity, which in turn reflects a weighted sum of factors. There thus exists a factor-level readout dimension, $\mathbf{b}^{\text{Fac}}$. Each weight in $\mathbf{b}^{\text{Fac}}$ describes the

net influence of one factor on the muscle. We describe the values of the factors, at time t , using a vector $\mathbf{x}(t)$. Using vector notation, the relevant weighted sum is $\mathbf{b}^{\text{FacT}} \mathbf{x}(t)$. (This can be shown formally by noting that $\mathbf{r}(t) \approx \mathbf{U} \mathbf{x}(t)$, where $\mathbf{U}$ captures each neuron's tuning for each factor. Setting $\mathbf{b}^{\text{FacT}} = \mathbf{b}^{\text{T}} \mathbf{U}$, we obtain $\mathbf{b}^{\text{T}} \mathbf{r}(t) \approx \mathbf{b}^{\text{FacT}} \mathbf{x}(t)$.) This weighted sum is predicted to remain constant during preparation.

This factor-level prediction is readily visualized in state space. Consider the model whose single-neuron responses are shown in

Fig. 3 | Illustration of the null-space hypothesis and its predictions.

a, Simulation of activity during a delayed-reach task, using a simplified version of the model in ref. 23. Six factors – two preparatory factors and four execution factors – obeyed rotational dynamics. Each neuron's firing rate was a weighted sum of those factors. To create preparatory activity, the two preparatory factors were endowed with non-zero values, specific to the upcoming reach. The duration of preparatory activity varied across reaches. Just before reach onset, preparatory factors decayed back to zero as their activity rotated into the four execution factors. Firing rates are shown, for 25 neurons, for three reaches in succession. Rates of 5,000 such neurons (grey circles) determined the activity of two muscles via weights in readout dimensions $\mathbf{b}_{\text{inactive}}$ and $\mathbf{b}_{\text{active}}$. The first muscle is largely inactive for all reaches (top red trace). The second muscle is active during execution but not preparation (bottom red trace), emulating what occurs in a typical delayed-reach task. **b**, Relationship between readout weights (dimensions $\mathbf{b}_{\text{inactive}}$ and $\mathbf{b}_{\text{active}}$) and factor-tuning weights (dimensions $\mathbf{u}_{1,6}$). The angle between a readout dimension (red) and a factor-tuning dimension (black) indicates their alignment, assessed via correlation of their elements: $\mathbf{b}^T \mathbf{u}$. The plotted angle, $\cos^{-1} \mathbf{b}^T \mathbf{u}$, is 90° if dimensions are orthogonal (if $\mathbf{b}^T \mathbf{u} = 0$). Most comparisons are nearly orthogonal, indicating that the factor has no influence on that readout. Near-orthogonality is expected, unless one imposes a relationship. We employed random readout weights and factor-tuning weights were also random, except $\mathbf{u}_1$ and $\mathbf{u}_3$ were modified to correlate with $\mathbf{b}_{\text{active}}$ ($r \approx 0.5$). Consequently, only factors 1 and 3 influence the active muscle (right), and no factors influence the inactive muscle (left). **c**, In large networks, it is natural for a muscle's activity to differ from that of most neurons driving it. We analysed only

'driving neurons': those providing positive net drive to the active muscle. Specifically, we analysed neuron n only if $b_{\text{active}_n} > 0.5$. For each such neuron, we computed the correlation between its firing rate (across all simulated reaches) and the active muscle's activity. The resulting distribution was broad, with many negative values and a mean (black triangle) that was only modestly positive. Thus, there is only a weak tendency for neurons to resemble the muscle they drive. **d**, Simulated neural trajectories (from the model above) in state space, illustrating why preparatory factors do not influence the active muscle and why no factors influence the inactive muscle. The state space is defined by one preparatory factor and one execution factor. Trajectories (one for each of the three reaches) span 350 ms, beginning at the transition from preparation to execution. Because readout dimensions are known for the model, their projections ($\mathbf{b}_{\text{active}}^{\text{Fac}}$ and $\mathbf{b}_{\text{inactive}}^{\text{Fac}}$) can be computed directly (red arrows). $\mathbf{b}_{\text{active}}^{\text{Fac}}$ is aligned with the execution factor, such that the three trajectories have different, and time-varying, patterns of activity in that direction (curved traces). The projection of trajectories onto $\mathbf{b}_{\text{active}}^{\text{Fac}}$ accounts for the initial change in the active muscle: positive, negative and positive for reach 1, 2 and 3, respectively (compare with panel **a**). $\mathbf{b}_{\text{active}}^{\text{Fac}}$ is orthogonal to the preparatory factor, which thus has no impact on the muscle. The readout dimension for the inactive muscle, $\mathbf{b}_{\text{inactive}}^{\text{Fac}}$, has almost no projection into these two dimensions (that is, the dot product of $\mathbf{b}_{\text{inactive}}^{\text{Fac}}$ with the two factor-tuning dimensions is nearly zero). Thus, neither factor influences the inactive muscle. **e**, Similar state-space plot for motor-cortex data. Readout-dimension projections, $\mathbf{b}^{\text{Fac}}$, are not known and must be estimated. jPC, neural dimension estimated by the jPCA method. Panel **e** adapted from ref. 23, Springer Nature Ltd.

Fig. 3a. For each reach condition there exists a neural trajectory describing how the factors evolve with time (Fig. 3d). Three trajectories are shown, one for each reach, beginning at the moment when preparation ends (circles) and execution commences. Because we know $\mathbf{b}_{\text{active}}$, we can compute its factor-level counterpart, $\mathbf{b}_{\text{active}}^{\text{Fac}}$ (long red vector). Separation amongst the preparatory states (circles) occurs in a direction orthogonal to $\mathbf{b}_{\text{active}}^{\text{Fac}}$. Consequently, there is no impact on the muscle. Once execution begins, trajectories change in ways that partially align with $\mathbf{b}_{\text{active}}^{\text{Fac}}$. For example, trajectories for reaches 1 and 2 produce increasing and decreasing readouts, respectively. Consequently, the active muscle shows an initial activity increase for reach 1 and a decrease for reach 2 (Fig. 3a).

Kaufman et al.³⁰ found that motor-cortex factor trajectories exhibit this same geometry (Fig. 3e). Factor estimation employed the jPCA dimensionality reduction method²³. Two estimated factors are plotted for three conditions (Fig. 3e), with each condition corresponding to a different reach type. The empirical preparatory-period states (circles, one per condition) are arranged in a line, and would thus avoid impacting muscle activity if $\mathbf{b}^{\text{Fac}}$ were oriented as shown (dashed vector). To test that prediction, $\mathbf{b}^{\text{Fac}}$ was estimated by regressing muscle activity against the estimated factors. Multiple instances of $\mathbf{b}^{\text{Fac}}$ were identified, because multiple muscles were recorded. As predicted, the spread of preparatory-period neural states was nearly orthogonal to each $\mathbf{b}^{\text{Fac}}$. Put differently, if one uses the set of $\mathbf{b}^{\text{Fac}}$ (one per muscle) to define an output-potent space, preparatory-period activity is largely confined to the orthogonal null-space. Importantly, times during the preparatory period were not used to estimate $\mathbf{b}^{\text{Fac}}$; the estimated readout directions naturally ignored preparatory activity, even though they were not encouraged to do so.

The empirical data matched another prediction of network-style models: in many dimensions, including the two shown (Fig. 3e), neural trajectories simply rotate. The ordering of states during preparation (for example, medium purple, then dark, then light) is thus largely

preserved during execution. These observations are consistent with a well-behaved dynamical flow-field: if the current factor-state is known, one can largely predict its short-term evolution (that is, its derivative)²³. Conversely, one can predict what preparatory neural activity must have been, given neural activity during execution⁶. This property – well-behaved dynamics with a large rotational component – is not an inevitable feature of trajectory evolution^{23,114}; muscle trajectories are far more haphazard²³, as are trajectories in other cortical areas^{38,115}. Yet this property is typical of network models trained on reaching tasks. It is not universal – networks can find more complex solutions if one allows them to²⁴ – but is standardly observed when networks achieve 'good' noise-robust solutions. We return to this point below in the context of execution-related activity. Here we simply note one reason why rotations may be prominent in network solutions (and thus why they may be prominent in motor cortex). Rotations allow a pattern of neural activity to be established in output-null dimensions during preparation, then enter output-potent dimensions during execution. This provides a mechanism by which preparatory activity can avoid causing current muscle activity, yet have a predictable influence on muscle activity in the future. This could of course happen in other ways, but a rotation is one of the simplest and most straightforward.

The null-space during execution

Null-spaces, recurrent networks and single-neuron tuning

Since the 1960s, researchers had debated the nature of execution-epoch tuning in motor cortex^{15,19–21,46,48,49,53,91,116–118}. Traditionally, tuning described how neural responses represent movement parameters. For example, if v_x and v_y are horizontal and vertical hand velocity, one might adopt the model $\mathbf{r}(t) \approx \mathbf{u}_{x\text{vel}} v_x(t) + \mathbf{u}_{y\text{vel}} v_y(t)$. The weights contained in vectors $\mathbf{u}_{x\text{vel}}$ and $\mathbf{u}_{y\text{vel}}$ describe each neuron's velocity tuning and thus its preferred reach direction. The intuitive appeal of such representational models collided with the finding that a neuron's preferred direction typically differs between preparation and

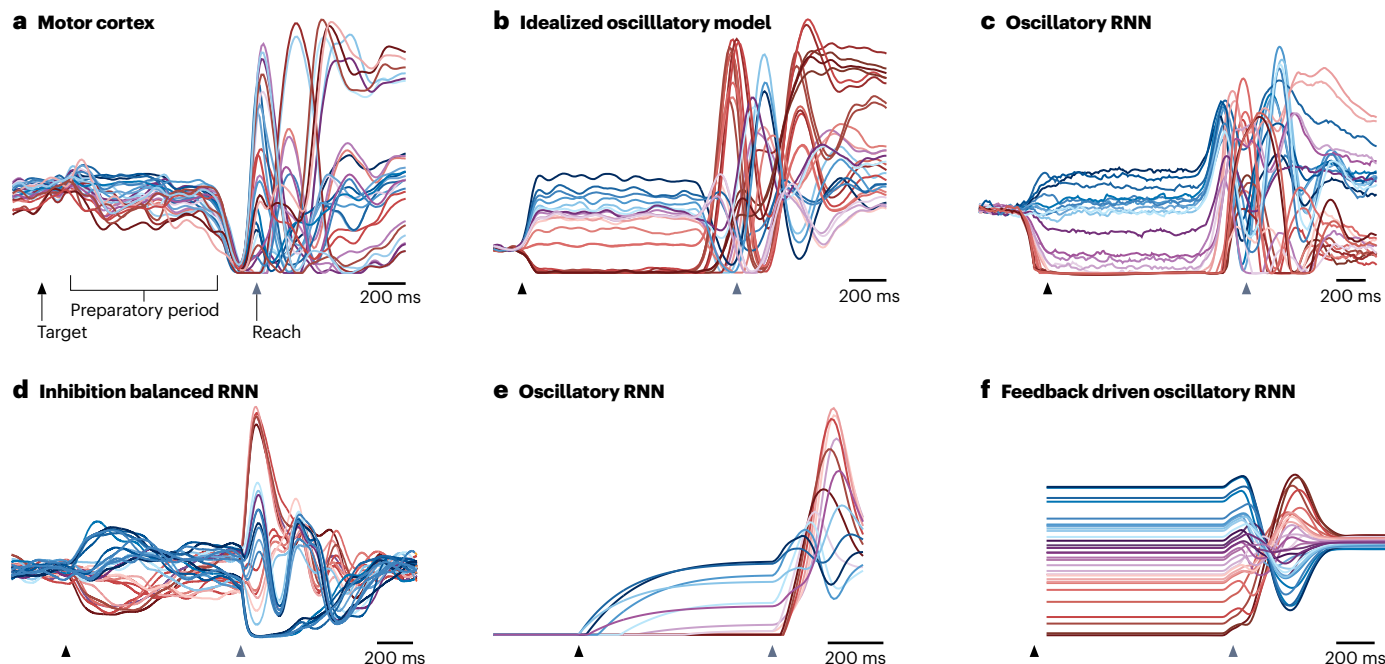


Fig. 4 | Single-neuron responses, during a delayed-reach task, from motor cortex and from simulated models. **a**, Response of a motor-cortex neuron recorded during the maze task. Each trace plots the trial-averaged firing rate for one behavioural condition (that is, one reach type). Preparatory-period activity is defined as activity after target onset (black arrow) and before the go cue (not shown). Execution-related activity begins ~ 150 ms before reach onset (grey arrow). Traces are colour-coded, from red to blue, based on the intensity of preparatory-period activity. **b–f**, The response of one example simulated neuron is plotted for each of five factor-dynamics models: one idealized factor-level model and four recurrent neural network (RNN) models. Colour-coding is as in panel **a** and is separate for each panel. In each of these models, single-neuron responses are a consequence of the factor-level dynamics used to generate motor outputs. Part of the appeal of this class of models was that single-neuron responses exhibit a variety of realistic features, despite never being fit to real neural responses. Two such features are illustrated by these examples. First, the tuning of preparatory-period activity (that is, the ordering with respect to condition) differs from that during

execution. Second, execution-related activity is often multiphasic, with no consistent tuning for condition. For example, consider the response of the simulated neuron in panel **b**. Blue-shaded traces correspond to conditions where the model generated muscle activity for down-and-left reaches. Thus, one would say that this neuron exhibits a down-and-left preference during the preparatory period. Early during execution, it exhibits the opposite preference: activity peaks highest for the red traces, corresponding to conditions where the model generated muscle activity for up-and-right reaches. After ~ 200 ms this preference has reversed (blue traces show the highest peaks) and ~ 300 ms later it reverses again. The other simulated neurons exhibit varying degrees of complexity, but all exhibit similar reversals. Data in panels **a** and **b** adapted from ref. 23, Springer Nature Ltd. Data in panel **c** adapted from ref. 24, Springer Nature Ltd. Data in panel **d** adapted with permission from ref. 108, Elsevier. Data in panel **e** adapted from ref. 110, CC BY 4.0 (<https://creativecommons.org/licenses/by/4.0/>). Data in panel **f** adapted with permission from ref. 121, CC BY 4.0 (<https://creativecommons.org/licenses/by/4.0/>).

execution^{6,14,103} (for example, Fig. 1b) and continues to be unstable during execution^{22,119,120} (for example, Fig. 4a).

While baffling from a traditional perspective^{19,20}, unstable preferred directions are standard in models^{6,23,24} that employ factor-state rotations and a null-space. Depending on the model, rotations reflect quasi-oscillatory dynamics^{23,24,110,121}, non-normal dynamics¹⁰⁸, or both⁸¹. Rotations create an orderly flow of activity from output-null to output-potent dimensions (as noted above) and generate the multiple peaks and valleys of muscle activity. Importantly, factor tuning is constant. Thus, if dynamics had been purely expansive (as in traditional ‘rising hill’ models) rather than rotational, each neuron would have displayed a consistent preferred direction¹⁰⁷. Yet as rotations unfold, the identity of the most-active factors changes. Many of those factors are output-null, and therefore do not correspond to any external parameter an experimenter might consider. Tuning thus appears inconstant when expressed with respect to external parameters.

Factor-based models with these features were found to produce realistic single-neuron responses (Fig. 4b–f) and rotations whose

frequency and duration were similar to those seen empirically (compare Fig. 3d and Fig. 3e). Models did so not because they were fit to data, but as a by-product of being trained to produce realistic outputs (for example, recorded patterns of muscle activity). Supporting the hypothesis that factor-state rotations involve output-null factors, orderly rotations are present in neural trajectories but not muscle trajectories^{23,115}.

Null-spaces and low tangling

Why use a null-space during execution? Why should neural trajectories not simply mirror the muscle trajectories they are intended to create? Observations during cycling tasks^{122,123}, where monkeys rhythmically cycle a hand-held pedal to move through a virtual environment, helped to craft a general answer. Consider a hypothetical pair of rhythmic patterns that might be produced by two output-potent factors (Fig. 5a). Without additional factors, the two-dimensional neural trajectory would be a figure-eight (Fig. 5b, top). A figure-eight trajectory crosses through the centre point twice, each time travelling

in a different direction. This situation – similar states with dissimilar derivatives – is referred to as ‘trajectory tangling’. Trajectory tangling is mechanistically problematic. If trajectories literally cross, they

cannot be generated by factor-dynamics (the definition of a dynamical system is that the state determines the derivative). Even if trajectories simply come close to crossing, the moment of high tangling creates a

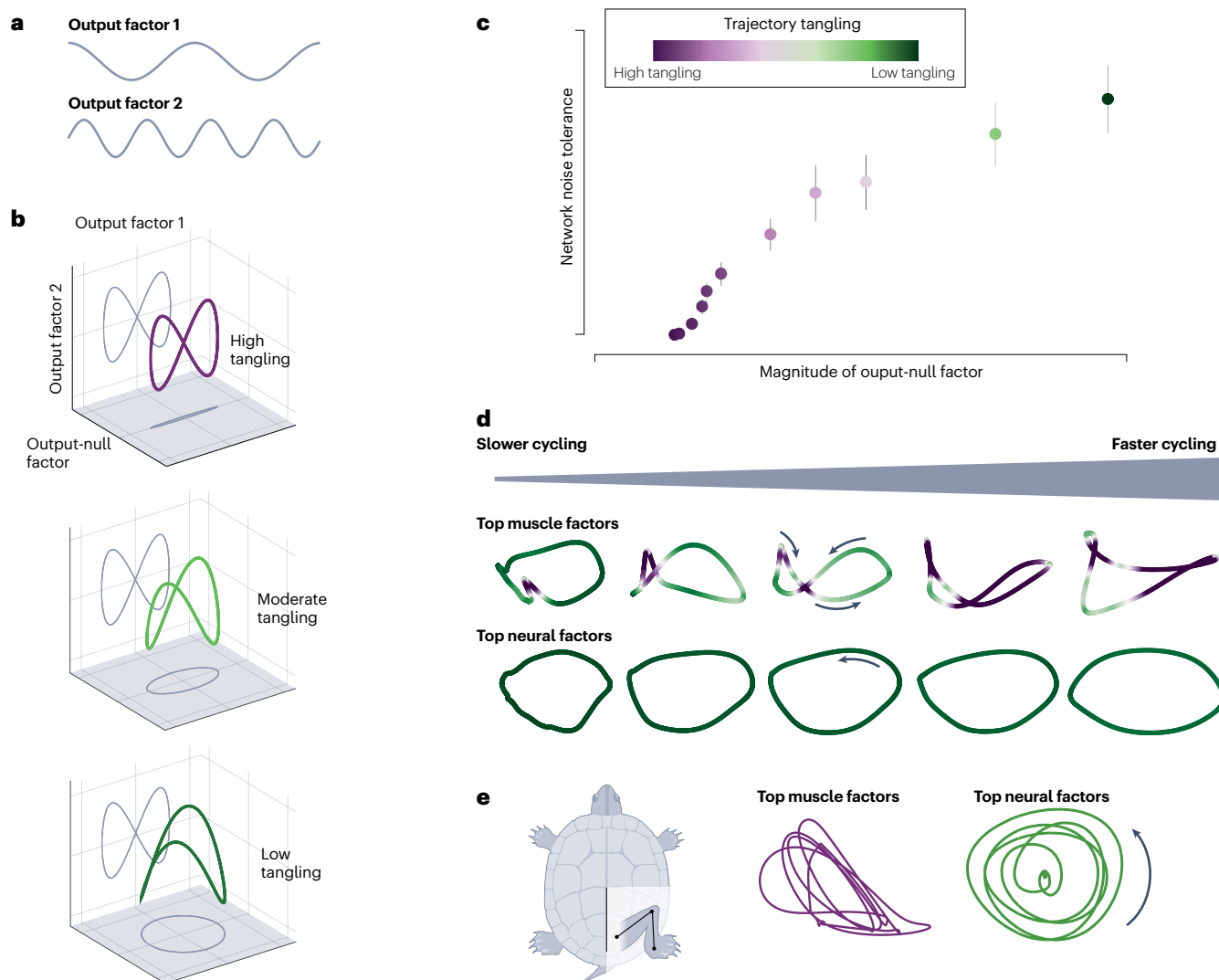


Fig. 5 | The null-space allows low trajectory tangling. **a**, Two hypothetical output factors during the cycling task. Each is an output-potent factor whose activity supplies one of the rhythmic activity patterns produced by the network. **b**, Three possible neural trajectories that might be observed in a network with the output factors from panel **a**. The grey figure-eight trajectory reflects the sinusoidal profiles, $\cos t$ and $\sin 2t$, of the output factors. Coloured trajectories are the ‘full’ neural trajectory of the network, including both output factors and an output-null factor described by $\sin t$. Trajectory tangling declines as the magnitude of the output-null factor increases from top to bottom. **c**, Noise-tolerance of artificial networks, trained to generate factor trajectories such as those in panel **b**, with different magnitudes of the output-null factor. Noise-tolerance improves as the magnitude of the output-null factor increases, owing to the reduction in trajectory tangling. **d**, Empirical data showing that motor-cortex trajectories tend to be far less tangled, and much more circular, than muscle-population trajectories. Muscle and neural trajectories are from a task where monkeys were trained to cycle a hand-held pedal at different speeds. Muscle activity was recorded from the major muscles of the arm and neural activity was recorded from motor cortex. Each subpanel corresponds to a given cycling speed (~ 0.8 Hz to ~ 2.1 Hz) and plots one cycle of a rhythmic trajectory. Trajectories were created by

projecting population activity (muscle or neural) onto its top two principal components (PCs). Each subpanel employs the top two PCs for that specific condition, to best capture that condition’s trajectory shape. Trace colour indicates the degree of tangling at each moment. Muscle trajectories were frequently tangled and took on a variety of shapes, some resembling the idealized figure-eight from panel **b**. Neural trajectories were consistently near-circular in these two dominant dimensions. Neural trajectories were non-circular in other dimensions, but the dominant circular trajectory ensured that tangling remained low. **e**, Low-tangled neural trajectories from a population of spinal interneurons. Data were collected during a rhythmic hindlimb scratching movement. Muscle trajectories (purple), computed from motor-nerve recordings, were more tangled and less circular than neural trajectories (green). Circular neural trajectories ran counter to the expectation of a push–pull pattern-generation mechanism. This highlights a theme: across multiple behaviours (reaching, cycling, scratching) neural trajectories are more circular than anticipated given traditional hypotheses, presumably because trending towards circular reduces trajectory tangling. Panel **a–c** adapted with permission from ref. 38, Elsevier. Panel **d** adapted from ref. 123, CC BY 4.0 (<https://creativecommons.org/licenses/by/4.0/>). Panel **e** adapted from ref. 126, Springer Nature Ltd.

Glossary

Single-neuron response

The response of a real (or simulated) neuron is notated as $r_n(t)$, where t indicates time and n indicates we are considering the n^{th} neuron.

Population response

The responses of N neurons, constituting an N -dimensional vector $\mathbf{r}(t)$. Each vector element, $r_n(t)$, describes the activity of one neuron.

Weight

A number indicating the degree to which one thing (for example, a neuron) influences something else (for example, another neuron). Weights come in many flavours and can summarize many things, including synaptic strengths and the impact of neural activity on muscle activity.

Neural dimension

An N -dimensional vector containing one weight per neuron. Just like the weights themselves, neural dimensions come in multiple flavours.

Weighted sum

Weighted sums of neural activity are central to many analyses that seek to understand network function (real or simulated). If the vector $\mathbf{v}$ is a neural dimension, one computes $\sum_{n=1}^N v_n r_n(t)$. Using vector notation, the weighted sum $\mathbf{a}(t) = \mathbf{v}^T \mathbf{r}(t)$ is referred to as ‘activity in dimension $\mathbf{v}$ ’.

Subspace

A set of one or more orthogonal neural dimensions. Dimensions are collected into columns of a matrix $\mathbf{V}$. The vector of weighted sums $\mathbf{a}(t) = \mathbf{V}^T \mathbf{r}(t)$ is referred to as ‘activity in the subspace of $\mathbf{V}$ ’.

Activity subspace

A subspace that fully captures neural responses. Suppose we computed $\mathbf{a}(t)$ as above (see subspaces). If we can successfully invert this relationship using $\mathbf{r}(t) = \mathbf{V} \mathbf{a}(t)$, then the dimensions in $\mathbf{V}$ span the activity subspace.

Dimensionality reduction

A method, such as principal component analysis (PCA), for estimating dimensions that constitute the activity subspace.

Readout

A signal that exits the neural population, such as a descending command for muscle activity (see Fig. 1c). Assuming linearity, the readout is a weighted sum of population responses: $\mathbf{z}(t) = \mathbf{b}^T \mathbf{r}(t)$ (see readout dimensions for how $\mathbf{b}$ is defined).

Readout dimensions

Each readout has an associated neural dimension, $\mathbf{b}$, containing weights (one per neuron) used for the readout. Analysis and visualization often employ a lower-dimensional $\mathbf{b}^{\text{fac}}$, with one weight per factor.

Output-potent space

The subspace, spanned by the readout dimensions, where neural activity impacts those readouts. Dimension $\mathbf{v}$ is ‘output-potent’ if it overlaps with a readout dimension: $\mathbf{b}^T \mathbf{v} \neq 0$.

Null-space

The subspace, orthogonal to the readout dimensions, where neural activity has no impact on those readouts. Dimension $\mathbf{v}$ is ‘output-null’ if it is orthogonal to each readout dimension: $\mathbf{b}^T \mathbf{v} = 0$.

Factors

For many recurrent networks, computation can be summarized by considering K factors rather than N neurons (see Fig. 1c), with $K \ll N$. Each factor is a weighted sum of activity: $x_i(t) = \mathbf{w}_i^T \mathbf{r}(t)$, where $\mathbf{w}_i$ is a neural dimension. Considering all factors: $\mathbf{x}(t) = \mathbf{W}^T \mathbf{r}(t)$. $\mathbf{W}$ reflects network connectivity (see recurrent connectivity).

Neural state

The value of the factors, $\mathbf{x}(t)$, at a particular moment t . ‘Neural state’ may also refer to the population response, $\mathbf{r}(t)$. Assuming factors are a valid summary of the population response, these definitions are equivalent.

State space

A method for visualizing the neural state as a point in a space where each axis corresponds to a factor (or some other weighted sum of neural activity).

Neural trajectory

The value of the factors, $\mathbf{x}(t)$, across a span of time. When plotted in state space, trajectories form traces that describe how activity changes with time (see Fig. 1c, inset).

Manifold

The shape formed by the collection of neural trajectories, defining the possible locations of the neural state within the activity subspace. For example, when cycling at different speeds, the manifold has a tube-like shape (see Fig. 2d).

Tuning

A description of how a neuron’s response reflects a specific quantity, such as a stimulus. In recurrent networks, factors can be the relevant quantities.

Factor tuning

The weight u_{ni} quantifies how the i^{th} factor impacts the n^{th} neuron. The neural dimension $\mathbf{u}_i$ quantifies the impact on all neurons. Given a matrix $\mathbf{U}$ of factor-tuning dimensions, population activity is $\mathbf{r}(t) \approx \mathbf{U} \mathbf{x}(t)$. The approximation reflects non-linearity and spiking variability. $\mathbf{U}$ reflects network connectivity (see recurrent connectivity), and defines the activity subspace.

Output-potent factors

Factor $x_i(t)$ is output-potent (that is, impacts the readout) if its factor-tuning dimension, $\mathbf{u}_i$, is output-potent (overlaps with one or more readout dimensions).

Output-null factors

Factor $x_i(t)$ is output-null (that is, does not impact the readout) if its factor-tuning dimension, $\mathbf{u}_i$, is output-null (lies within the null-space).

Recurrent connectivity

Synaptic connections that cause activity to flow in ‘loops’. The causal flow is simplified by considering

factors (Fig. 1c) and matrices $\mathbf{W}$ and $\mathbf{U}$ (see factors and factor tuning). Factors are weighted sums of neural responses: $\mathbf{x}(t) = \mathbf{W}^T \mathbf{r}(t)$. Responses reflect the factors, $\mathbf{r}(t) \approx \mathbf{U} \mathbf{x}(t)$, completing the loop. In some networks, $\mathbf{W}$ and $\mathbf{U}$ can be computed directly from synaptic connectivity.

Factor-level dynamics

In recurrent networks, activity at the present moment drives activity at the next moment. Such dynamics are often best described at the factor level: $\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}) + \mathbf{y}$. The function $\mathbf{f}$ defines a flow-field in state space (see Fig. 1c, inset). $\mathbf{y}$ captures inputs from outside the observed system.

Rotational dynamics

Dynamics linking two factors, x_1 and x_2 , such that neural states flow from one to the other while preserving their order (see Fig. 2a). Suppose that, for three conditions during preparation, factor x_{prep} takes values $\{-1, 1, 2\}$ while factor x_{exec} takes values $\{0, 0, 0\}$. After a 90° rotation, activity has left the preparatory subspace, $x_{\text{prep}} = \{0, 0, 0\}$, and entered the execution subspace, $x_{\text{exec}} = \{-1, 1, 2\}$. Depending on the network, rotations might continue or might end after -90°.

Factor estimation

To estimate factors from measurements of neural activity alone, without knowledge of connectivity, experimenters use dimensionality reduction to estimate the activity subspace and invert the relationship $\mathbf{r}(t) = \mathbf{U} \mathbf{x}(t)$. For example, one may use $\mathbf{x}(t) = \mathbf{U}^T \mathbf{r}(t)$, assuming the estimated $\mathbf{U}$ is orthonormal.

Motor neurons

Neurons in the spinal cord that connect directly to a muscle. Each motor neuron spike produces one spike in the muscle fibres it innervates. Motor neurons receive direct (monosynaptic) and indirect (polysynaptic) inputs from motor-cortex neurons.

potential instability where noise or small perturbations could disrupt the future neural trajectory³⁸ and the resulting behaviour. Adding an output-null factor reduces tangling, resulting in a more circular trajectory (Fig. 5b, bottom) and much-improved noise robustness (Fig. 5c). This illustrates that networks perform best if activity does not simply mirror the outputs being produced, but reflects additional output-null factors.

Thus, a strong prediction of the factor-dynamics hypothesis is low neural-trajectory tangling enabled by output-null factors. As predicted, motor-cortex neural trajectories display dramatically lower tangling than muscle trajectories, across a variety of tasks^{38,123}. For example, when cycling at different speeds, muscle-state trajectories traced a variety of tangled shapes, including some resembling a figure-eight (Fig. 5d, top). By contrast, neural trajectories were consistently close to circular in the dominant dimensions (Fig. 5d, bottom). Although neural trajectories do not resemble muscle trajectories, muscle activity can be accurately decoded by considering all factors^{38,123}. Thus, neural trajectories ‘contain’ muscle trajectories, but also reflect output-null factors that reduce tangling.

Although strategies for minimizing tangling are task-specific, a common motif is for trajectories to trend towards circular^{38,123,124}. Circular neural trajectories are also observed in primate motor cortex during locomotion¹²⁴, and in robot agents guided by neural-network controllers¹²⁵. Rotations during reaching can be seen as an instance of this general strategy²⁴. Circular neural trajectories were also observed in turtle spinal-interneuron populations during generation of rhythmic motor patterns¹²⁶ (Fig. 5e), a finding that contrasted with the traditional expectation of alternating flexor-controlling and extensor-controlling subpopulations. The resulting change in perspective paralleled that during reaching. In both cases, early hypotheses naturally tended to focus on aspects of neural activity that matched motor outputs (for example, a muscle-driving burst). As the range of plausible network strategies became better understood, hypotheses began to include output-null factors and became better able to explain neural activity.

Importantly, low tangling is desirable only when the present factor-state must determine the future factor-state. There are thus moments where trajectory tangling should be high – for example, when reach-target onset first creates preparatory activity – and indeed this is the case. Tangling is also high in sensory areas, consistent with neural activity being dominated by inputs³⁸. Interestingly, trajectory tangling is high in motor cortex during grasping, arguing that the shaping of the digits is driven by continuous guiding inputs external to the dynamics observed in motor cortex¹²⁷.

A subtle point is that the null-space hypothesis concerns a general network mechanism, not a physical location. It remains unknown how much null-space activity is lost at each anatomical stage, from the general motor cortex population to cortico-spinal neurons, to spinal interneurons and then motor neurons. It is likely that activity becomes more ‘muscle-like’ at each stage^{38,102,128}. It is also worth stressing that the idea of motor commands descending a hierarchy – working their way from motor cortex to spinal interneurons and then motor neurons – is overly simplistic. As highlighted by the results in turtle, spinal-interneuron populations can have strong factor-level dynamics of their own. Additionally, cortex forms recurrent loops with the spinal cord, basal ganglia, cerebellum and thalamus^{96,97,129,130}. Thus, when one observes factor-level dynamics in cortex, one should not presume that their origin lies principally within cortical recurrent connectivity.

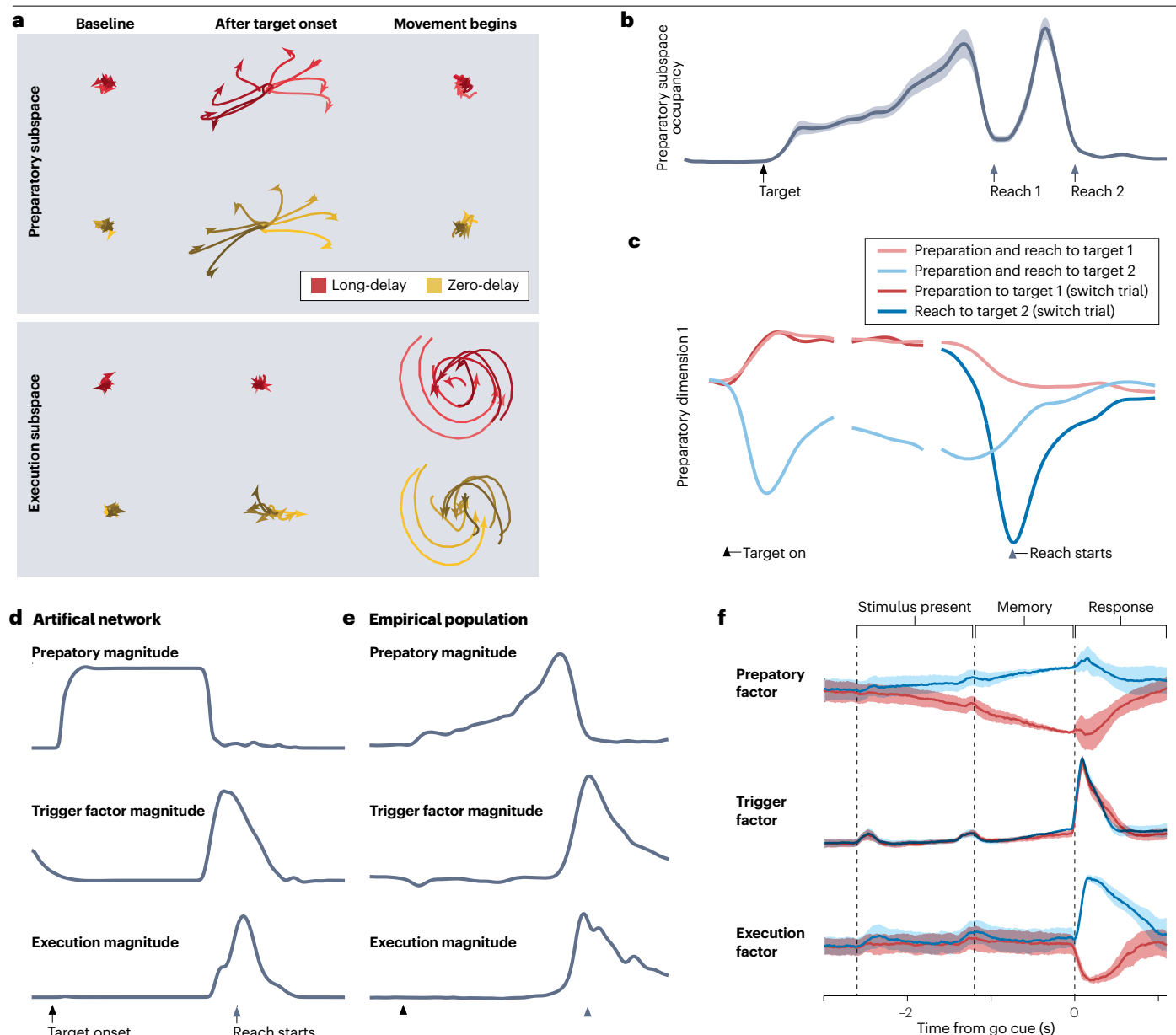
Thinking in terms of null-spaces generates predictions and aids analysis

Preparation and execution involve distinct factors

Early studies of motor-cortex preparatory activity (for example, ref. 2) often categorized neurons as preparatory versus execution-related, yet also documented that both types of activity can occur in the same neuron. The tendency to categorize was motivated both by convenience (neurons with similar properties could be analysed together) and by the expectation that neural signals with different ‘meaning’ might be separated into distinct neural populations. This expectation holds in many contexts, and even to some degree in motor cortex, where neurons deep in the sulcus are typically dominated by execution-related activity. Anatomical separation is presumably beneficial when signals should not interact; one presumes it is important for execution-related activity to become, at some stage, independent of preparatory activity.

Yet signals often must interact, and the most natural way for output-null factors to influence output-potent factors is for them to be mixed within the same neurons. For this reason, the emergence of the factor-dynamics framework was intertwined with the field’s increasing appreciation that mixed responses are common¹³¹ and might be computationally ‘good’ rather than a by-product of biological messiness. Factor-dynamics models yielded realistically diverse single-neuron responses²², often modelled using random combinations of factors²³. Single-neuron responses in motor cortex were found to reflect putative output-null and output-potent factors in roughly random proportions³⁰. Random (‘category-free’)¹³², or nearly random¹³³, tuning was observed in other cortical areas during other tasks. Yet although mixing may be computationally beneficial, it can make single-neuron responses difficult to interpret. Researchers thus pursued the goal of ‘de-mixing’ population activity into interpretable factors^{131,132}.

In motor cortex, an unexpected finding – and one that greatly aided analysis – was that entirely distinct sets of factors are active during preparation and execution^{134,135}. At first it had been expected that output-null factors, once active during preparation, would continue to be active during execution^{24,30}. Yet essentially all factors that are active during preparation fall silent during execution. Other factors (presumably both output-potent and output-null) become active only during execution (Fig. 2a). Formally, if neural dimensions $\mathbf{u}_{\text{prep}}$ and $\mathbf{u}_{\text{exec}}$ describe tuning for one preparation-active factor and one execution-active factor, those dimensions are orthogonal: $\mathbf{u}_{\text{prep}}^T \mathbf{u}_{\text{exec}} \approx 0$. A hypothesized benefit of orthogonality is that the dynamics linking preparatory and execution factors are more readily modulated, allowing preparation to be refined without engaging non-normal execution dynamics¹⁰⁹, and allowing preparation for the next movement to avoid perturbing the current movement⁸¹. Regardless of the benefits to the network, orthogonality is very beneficial to the experimenter. If one can estimate $\mathbf{u}_{\text{prep}}$ and $\mathbf{u}_{\text{exec}}$, one can isolate preparation-related and execution-related activity by projecting the population response onto these dimensions, using the weighted sums $\mathbf{u}_{\text{prep}}^T \mathbf{r}(t)$ and $\mathbf{u}_{\text{exec}}^T \mathbf{r}(t)$. This allowed experimenters to examine preparatory activity based on the relevant dimension, rather than based on a temporal epoch. In practice, there exist multiple dimensions of each type (multiple $\mathbf{u}_{\text{prep}}$ and $\mathbf{u}_{\text{exec}}$), forming distinct preparatory and execution-related subspaces. Studies thus compute ‘activity in the preparatory subspace’ (the estimated activity of preparatory factors) and ‘activity in the execution subspace’ (the estimated activity of execution factors).



This approach made it possible to revisit long-standing questions. Is preparation essential or a luxury afforded by a delay? Might preparation be skipped under time pressure? Might preparatory activity actually be an artefact of movement suppression during the artificially imposed delay? To address these topics, prior studies had compared pre-movement activity (on zero-delay trials) with preparatory-period activity (on trials with a delay). If preparation is essential, pre-movement activity should be a blend of preparation-related and execution-related activity, with preparation-related activity arising first. In agreement with this, pre-movement activity and preparatory-period activity share qualitative similarities^{9,14}. Yet quantitative differences exist, suggesting preparation might be skipped¹³⁶. This uncertainty was resolved using the approach described above. Preparatory and execution subspaces were estimated using trials with an imposed preparatory period (long-delay trials). Activity in those subspaces was then

examined for trials with no imposed preparatory period (zero-delay trials). The same set of preparatory-factor events, and the same set of execution-factor events, was found to occur with or without a delay¹³⁵ (Fig. 6a). Critically, this remained true even when reaction times were very short, consistent with the proposal that preparation is an essential precursor to voluntary movement^{9,14,137,138} and cannot be skipped.

Another long-standing question regarded whether, for rapid highly-practised sequences, the full sequence might be prepared in advance and executed holistically. This holistic hypothesis predicted one bout of preparation followed by one (long) bout of execution. Alternatively, each movement might be prepared and executed independently. Surprisingly, the latter was observed: during two-reach sequences the preparatory subspace was occupied twice (Fig. 6b) and execution-related events similarly repeated⁸¹. This was true even for very rapid sequences. These observations ran counter to the expectation

Fig. 6 | The ability to estimate interpretable factors allowed experiments to address key questions regarding motor cortex. **a**, Experiment addressing whether preparatory activity occurs without an experimenter-imposed delay. Long-delay and zero-delay trials were cued by different stimuli. Long-delay trials allowed estimation of factor-tuning dimensions for putative preparatory and execution factors. The relevant subspaces, $\mathbf{U}_{\text{prep}}$ and $\mathbf{U}_{\text{exec}}$, were nearly orthogonal: $\mathbf{U}_{\text{prep}}^T \mathbf{U}_{\text{exec}} \approx 0$. To estimate factors during zero-delay trials, the population response was projected into both subspaces, using $\mathbf{U}_{\text{prep}}^T \mathbf{r}(t)$ and $\mathbf{U}_{\text{exec}}^T \mathbf{r}(t)$. Trajectories are plotted for two estimated preparatory factors (top) and two estimated execution factors (bottom), across eight reach directions. Trajectories span 150 ms, beginning 150 ms before target onset (left), at target onset (middle) and 50 ms after movement onset (right). Preparatory-factor trajectories are similar for long-delay (red) and zero-delay (yellow) trials. Execution-factor trajectories also behave similarly. Thus, reaching involves the same preparatory-factor and execution-factor events, regardless of the presence of an imposed delay. **b**, Experiment addressing whether a sequence of multiple movements employs one preparatory phase per movement or one preparatory phase per sequence. A delayed-reach task employed single-reach and double-reach trials. Single-reach trials were used to estimate the preparatory subspace, $\mathbf{U}_{\text{prep}}$. To estimate preparatory factors during double-reach trials, the population response was projected into this subspace, using $\mathbf{U}_{\text{prep}}^T \mathbf{r}(t)$. The magnitude of activity, across all preparatory factors, is plotted with standard error. Preparatory factors become active during the preparatory period (from target onset until first-reach onset; arrows) in the usual way. Preparatory factors then become active again before second-reach onset. Thus, rapid two-reach sequences involve one preparatory phase per reach. **c**, Experiment addressing whether last-moment corrections involve re-preparation. Using the same strategy as above, factor-tuning dimensions were estimated using data from trials following the standard instructed-delay paradigm. Estimated factors were then examined on trials where the target suddenly switched to a new location soon after the go cue. One preparatory factor is shown for two non-switch conditions (light red and light blue for target location one and two) and for one condition where the target switched from location one to location two.

Before the switch (red), the preparatory factor was appropriate for a reach to target one. After the switch (blue), the preparatory factor swiftly changes to become appropriate for target two. These findings support the hypothesis that it is important for preparatory activity to be appropriate for the intended target by the time the reach begins. They also demonstrate that, although preparation often appears static in instructed-delay tasks, it is in fact highly flexible and can change quickly to track the intended target. **d**, Simulation illustrating the prepare–trigger–execute motif. A recurrent network was trained to generate muscle activity for single-reach trials and double-reach trials, as in the experiment described in panel **b**. Data are shown only for single-reach trials. Following training, the network employed distinct preparatory factors (top, showing zero magnitude during reaching) and execution factors (bottom, showing zero magnitude during preparation), with the transition between preparation and execution mediated by trigger factors (middle). This clear separation did not emerge in networks trained only on single reaches; preparatory and execution factors became entirely distinct only when networks also had to generate rapid two-reach sequences. This suggests that distinct factors are a network strategy ensuring that preparatory and execution factors interact only briefly following the trigger, allowing preparation for the next movement to occur while the present movement completes (as in panel **b**). **e**, A similar prepare–trigger–execute motif, with distinct preparatory and execution factors, is present for empirical data recorded from non-human primates using the same task. **f**, Experiment addressing whether a similar prepare–trigger–execute motif occurs in mice performing an instructed-licking task. There are fewer factors of each type (essentially only one of each) but the overall motif is similar. Preparatory-factor activity arises following stimulus onset and is maintained during a memory period. Trigger-factor activity arises after a sensory go cue. Execution-factor activity swiftly follows. Red and blue traces correspond to trials where the mouse was instructed to lick left or right. Panel **a** adapted from ref. 135, CC BY 4.0 (<https://creativecommons.org/licenses/by/4.0/>). Panel **b** adapted from ref. 81, Springer Nature Ltd. Panel **c** adapted with permission from ref. 136, Elsevier. Panels **d** and **e** adapted from ref. 81, Springer Nature Ltd. Panel **f** adapted with permission from ref. 149, Elsevier.

that preparation was necessarily a time-consuming process^{8,914,137,139} that would slow sequence production¹⁴⁰. Instead, preparation for the next reach occurred swiftly, just as the current reach was finishing.

A similar experimental strategy was leveraged to study the neural basis of rapid corrections. Preparatory-factor activity typically collapses (becomes similar across reaches) around movement onset (Fig. 6c, light traces). Yet when the intended target suddenly switches, preparatory-factor activity does not collapse but switches to reflect the new target (Fig. 6c, dark traces)¹⁴¹. This finding argued for a broader role for preparatory factors: responding to inputs (for example, those arriving from visual areas) whenever that information is needed to influence future movement. Consistent with this interpretation, correction-related activity first appears in output-null factors¹⁴².

Investigations of the prepare–trigger–execute motif

Early studies noted that “preparation for movement is a rather vague concept”^{8,143}. The above results made it possible to overcome this vagueness by defining preparatory activity not in terms of phenomenology (as activity that occurs at a certain time) but in terms of computational role. A working definition was that a factor is ‘preparatory’ if it responds to inputs, interacts with execution factors to shape execution, yet has no influence until a triggering input arrives. This separate triggering input was proposed^{9,23,68,144} as an alternative to the threshold-crossing hypothesis, and as a way of achieving the flexibility associated with voluntary movement. A separate trigger could allow preparation to

be extended or brief^{135,144}, affording opportunities for refinement¹⁴⁵, changes of mind¹⁴⁶ or cancellation¹⁴⁷.

The resulting prepare–trigger–execute hypothesis holds that one set of network inputs drives the preparatory factors and achieves a ‘good’ preparatory-factor state prior to movement onset. A subsequent input – effectively an internal go cue – drives distinct ‘trigger factors’. Trigger factors are themselves output-null. Yet by changing the location in factor-space and altering local factor-level dynamics (Fig. 2b), they allow preparatory factors to influence execution factors (Fig. 2a). Network simulations confirmed this strategy was effective²⁴, and generated multiple predictions regarding the putative trigger factors. Because they determine reach timing but not identity, trigger factors should behave similarly regardless of reach direction or type. Because they must profoundly alter local dynamics, trigger factors should cause a large overall change in neural activity. Because they serve as an internal go cue (not a sensory response), trigger factors should evolve identically regardless of how a movement is initiated. Indeed, trigger factors should anticipate, on each trial, the exact moment when reaching begins and should be locked to that moment rather than any sensory event. Last, because trigger factors are proposed to be output-null, they should be absent in muscle activity. All predictions were confirmed^{115,148}, solidifying the hypothesis that reaching employs a three-stage prepare–trigger–execute motif^{24,81,115,148}. Each stage involves distinct factors (Fig. 6d,e). Most of these – all preparatory factors, all trigger factors and many execution factors – are thought to be output-null.

The prepare–trigger–execute hypothesis is agnostic to many implementation details. It can be realized in spiking networks^{77,108}, rate-based networks^{24,81,110} and idealized factor-level models²³. Regardless of implementation, the hypothesis seeks to be mechanistic: it proposes a chain of events culminating in muscle activity. Experiments in mice leveraged this concrete causal chain to supply predictions regarding the impact of perturbations. Much of this work took place in the context of directional licking following an instructed-memory period. This task is not exactly analogous to delayed-reaching in primates, and neither is the key cortical area – anterolateral motor cortex – analogous to motor cortex. Yet much is preserved at the factor level and a prepare–trigger–execute motif is present¹⁴⁹ (Fig. 6f). Perturbations, with simultaneous factor-level observations, revealed that preparatory dynamics are surprisingly robust¹² and reflect both cortical recurrence and loops involving the thalamus¹² and cerebellum¹⁵⁰. Owing to this robustness, preparatory activity recovers, and behaviour remains accurate, following unilateral optogenetic perturbations of anterolateral motor cortex¹¹. Following bilateral perturbations, preparatory activity loses its link to the stimulus but still predicts lick direction¹¹, consistent with the hypothesis that preparatory activity determines pending movement^{1,9,151}. To test the hypothesis that the trigger factor engages execution dynamics, its origins were traced and manipulated¹⁴⁹. Optogenetically inducing the trigger factor induced movement initiation. Suppressing the trigger factor prevented initiation. Inducing the trigger factor generated licking only in the presence of preparatory activity, consistent with the hypothesis that the trigger factor is output-null, and creates movement by allowing preparatory factors to influence execution factors.

These results extended and complemented an earlier rodent-reaching study where the motor-cortex factor trajectory, and accompanying movement, were interrupted by optogenetic perturbation of thalamic activity⁹⁷. This finding supported the hypothesis that cortically observed dynamics depend on cortico-thalamic recurrence. Additionally (or alternatively), thalamic perturbations may disrupt thalamic inputs driving the trigger factors; the top two principal components (PCs) of cortical trajectories, which typically reflect trigger factors, were strongly disrupted by the perturbation. In a related experiment, inhibiting thalamic activity during a lever-push task blocked initiation, whereas inducing it triggered initiation, demonstrating a likely role of thalamic activity in triggering movement¹²⁹. The evoked movement was context-dependent, consistent with the idea that movement identity depends on the preparatory state, not the trigger itself.

These experiments demonstrate that cellular-level perturbations can alter, in interpretable ways, factor-level computational mechanisms. Indeed, perturbation-induced effects often made sense only when interpreted using factor-level hypotheses^{97,149}. This highlights that the goal should not be to find a single level of mechanistic description, but to employ a framework that links cellular and population-level properties, making it possible to move amongst levels depending on the question being asked and the tools available for answering it.

Future directions, open questions and controversies

Flexibility-via-subspace hypothesis

The finding that distinct factors are active during preparation and execution (discussed above) suggested a broader possibility: a circuit might use different active factors, governed by different dynamics, to flexibly switch amongst computations¹³⁴. Because different factors are associated with different factor-tuning dimensions (different u_i),

different computations would occur in distinct, orthogonal, subspaces (Fig. 2e). Evidence for this ‘flexibility-via-subspace’ hypothesis was observed in experiments where mice performed two distinct tasks in the same session: walking and lever-pulling¹⁵² (Fig. 7a) or co-contraction versus alternation¹⁵³ (Fig. 7b). The presence of behaviour-specific factors helped to explain why optogenetic silencing impaired lever-pulling but not walking; this would be expected if activity in the vertical dimension (Fig. 7a) was output-potent whereas that in the horizontal dimension was output-null.

These findings related to hypotheses regarding multifunctional circuits¹⁵⁴, and to the finding that motor-cortex responses change in seemingly random ways between reaching and posture tasks¹⁵⁵, consistent with different factors being used. The flexibility-via-subspace hypothesis also helped make sense of the finding that motor-cortex activity is surprisingly bilateral. Although neurons in left motor cortex control the right arm, they are robustly active when the left arm moves and the right arm remains stationary^{106,112,156–158}. Why does this extensive left-hemisphere neural activity not impact right-arm muscles? The factor-dynamics framework suggested a simple hypothesis: distinct sets of left-arm and right-arm factors, yielding orthogonal left-arm and right-arm subspaces. If so, readouts sent to the right arm, derived from right-arm factors, would naturally ignore left-arm factors. No special ‘ipsilateral activity suppressing’ mechanism would be needed, just as no special ‘preparation suppressing mechanism’ is needed. Orthogonal left-arm and right-arm subspaces were confirmed both when countering loads¹⁰⁶ (Fig. 7c) and during cycling¹¹² (Fig. 7d).

Network simulations indicate that flexibility-via-subspace is a valid strategy. Networks can flexibly perform multiple cognitive tasks by recombining distinct motifs involving distinct factors^{159,160}. Similarly, networks can produce a library of motor motifs by switching active factors between motifs¹⁶¹. When networks do so, different computations occur in orthogonal subspaces. For example, DePasquale et al. trained a spiking network that performs two entirely unrelated tasks – cycling and reaching – using distinct factors⁷⁷. One subspace is active during cycling (Fig. 7e, top) and a completely different subspace is active during reaching (Fig. 7e, bottom). This strategy allows one network to perform two completely distinct computations. A more continuous version of this strategy (Fig. 2d) appears when a single task requires variations upon a given skill. When cycling at different speeds¹²³, and when reaching in different directions¹⁶², factor trajectories are separated both by a null-space translation and by ‘tilting’ trajectories into increasingly different subspaces, a strategy predicted by network solutions¹²³.

The possibility that each task uses a subset of all factors may explain why considerable neural variance is often accounted for by a moderate number of dimensions^{23,163–165}. Each task may use a small percentage of all available factors. At the same time, the nervous system may reuse computations (and thus factors) when possible^{159,160,166}, limiting the total number of occupied dimensions. How this tension plays out is likely to be a theme of future research. It also highlights an older tension. Motor-cortex activity is low-dimensional in the practical sense that dimensionality reduction is useful for estimating the dominant factors and testing competing predictions. Presumably it is also true that neurons greatly outnumber factors. Yet the adoption of dimensionality reduction techniques was not motivated by any belief that motor-cortex activity is unusually low-dimensional. Quite the opposite: dimensionality reduction was adopted because motor-cortex responses were far more heterogeneous and high-dimensional than predicted by traditional representational hypotheses^{22,76}.

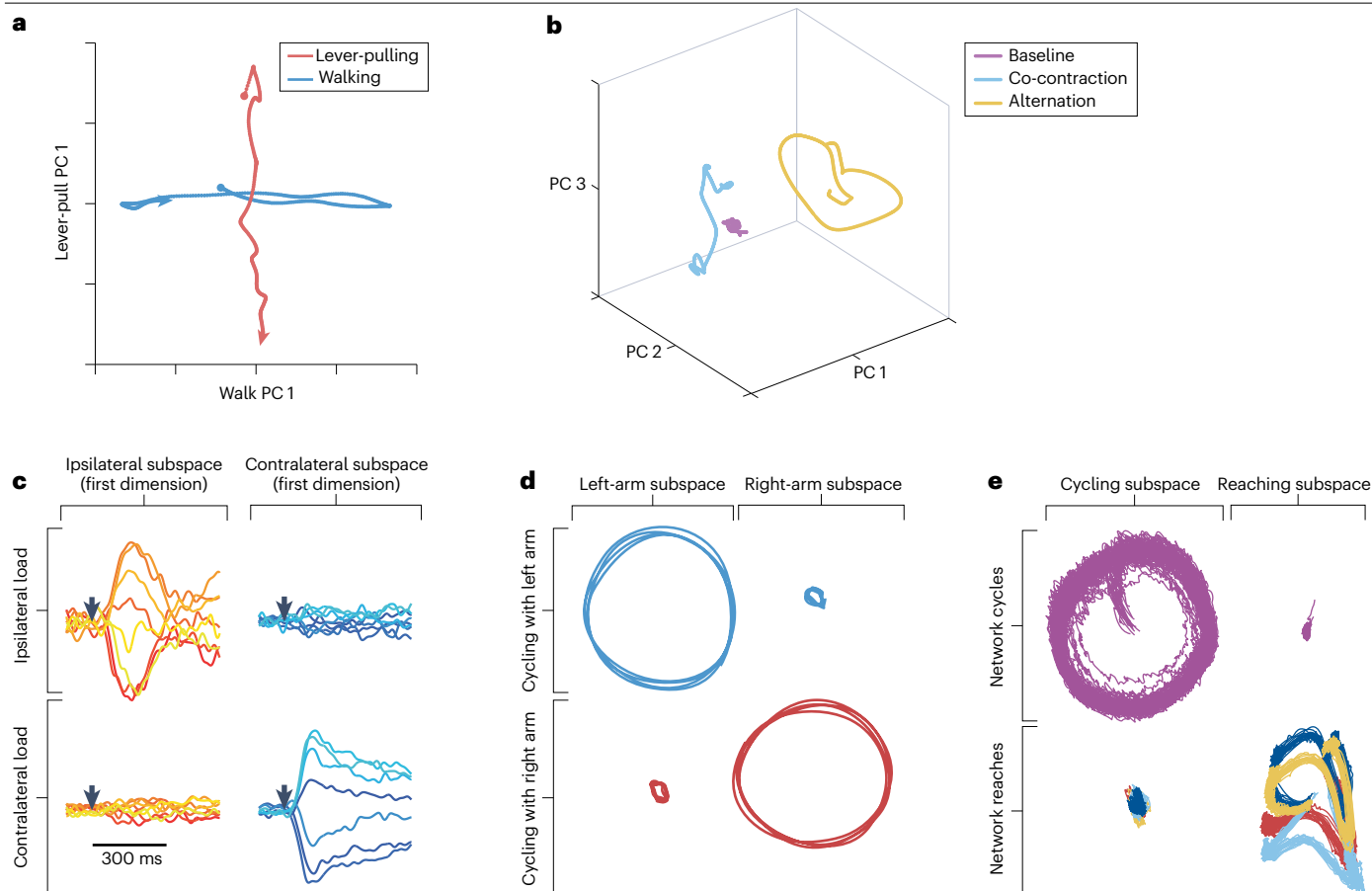


Fig. 7 | Data and simulations supporting the flexibility-via-subspace hypothesis. **a**, Mouse motor-cortex neural trajectories during walking and lever-pulling tasks. Each axis captures population activity in a dimension derived from the top principal component (PC) of population activity during one task. The population response during walking (blue) is restricted to the horizontal axis, whereas that during the lever-pull (red) is restricted to the vertical axis. Abolishing the blue trajectory (via optogenetic silencing during walking) had a small/delayed impact on behaviour, whereas abolishing the red trajectory (via optogenetic silencing during lever-pulling) had a large/immediate impact. These findings argue that walking and lever-pulling use distinct factors, with only the lever-pulling factors being output-potent. **b**, Mouse motor-cortex neural trajectories during co-contraction and alternation tasks. Each axis captures activity in one PC of the population response across both tasks. The roughly linear trajectory during co-contraction (blue) is largely orthogonal to the circular trajectory during alternation (yellow), and also separated from it by a translation. These findings support the hypothesis that different factors, and different regions of neural state space, are used when performing different tasks. **c**, Monkey motor-cortex activity in two orthogonal neural dimensions, during a task where the onset of a load is countered with the ipsilateral or contralateral arm. Left: activity in an 'ipsilateral-subspace dimension', constructed to capture population activity when using the ipsilateral arm. Right: activity in a

'contralateral-subspace dimension', constructed to capture population activity when using the contralateral arm. The two rows show activity when countering loads applied to ipsilateral (top) or contralateral (bottom) arms. Trace colours correspond to different load directions. Activity, following load onset, is largely restricted to just one of the two neural dimensions, supporting the hypothesis of distinct left-arm and right-arm factors. **d**, Monkey motor-cortex neural trajectories in two orthogonal neural subspaces, when cycling a hand-held pedal. The two rows show activity when cycling with the left arm (top, blue) and right arm (bottom, red). Activity when using the left arm is restricted to a left-arm subspace (left column). Activity when using the right arm is restricted to a right-arm subspace (right column). As in panel **c**, these results indicate distinct left-arm and right-arm factors. **e**, Neural trajectories for a recurrent spiking network that generates deltoid muscle activity during reaching and cycling tasks. The two rows show activity when cycling (top, purple) and when reaching in different directions (bottom, multiple colours). Cycling-task activity is restricted to a cycling subspace (left column). Reach-task activity is restricted to a reaching subspace (right column). Panel **a** adapted with permission from ref. 152, Elsevier. Panel **b** adapted with permission from ref. 153, Elsevier. Panel **c** adapted from ref. 106, CC BY 4.0 (<https://creativecommons.org/licenses/by/4.0/>). Panel **d** adapted with permission from ref. 112, CC BY 4.0 (<https://creativecommons.org/licenses/by/4.0/>). Panel **e** adapted with permission from ref. 77, Elsevier.

Those heterogeneous responses were poorly summarized by traditional methods such as the population-vector or population PSTH (an average based on every neuron's response during its preferred condition). Methods such as PCA provided an appealing alternative because they could use as many dimensions as needed to describe

the data. Using such methods, dimensionality has been estimated as dozens²² to hundreds³⁹. As predicted by factor-dynamics models (and in opposition to representational models), dimensionality is much higher when measured across neurons versus across behavioural conditions⁷⁶, and approaches smoothness-based limits^{167,168}.

Dimensionality estimates are likely to grow as it becomes possible to use multiple tasks in the same session. Yet although the activity subspace may be high-dimensional, most regions of this subspace are likely to remain unoccupied¹⁶⁹. For example, the data shown in Fig. 2d form a tube whose middle is never visited. In this example, understanding network dynamics (and phenomena such as trajectory tangling) requires considering all three factors. Yet links to behaviour can benefit from considering just two quantities: the phase and height on the tube-shaped manifold correspond, roughly, to the cycling phase and angular velocity. Facilitating such links to behaviour is a common goal of non-linear dimensionality reduction^{170–172}, which attempts to describe neural manifolds using as few dimensions as possible. In the present example, the difference between three (linear) dimensions and two (non-linear) dimensions may seem unremarkable. Yet when many factors are used across many tasks, manifold geometry threatens to become quite complex^{164,169}. In such cases, one may wish to use both linear dimensionality reduction – to estimate factors and describe the dynamics that subserve computation – and non-linear dimensionality reduction – to estimate the shape of the neural manifold and parameterize activity more efficiently.

If much is output-null, what signals are output-potent?

Early in the study of motor cortex, a natural hope was that the downstream parameters being controlled might be inferred from single-neuron responses^{116,117}. For example, single-neuron responses might resemble force, muscle activity or hand velocity. Yet with an expansive null-space, single-neuron responses rarely resemble outputs even when they drive them directly (as illustrated by the simulations shown in Fig. 3a–c), and there is no simple recipe for determining which factors are output-potent. Despite such challenges, a tentative consensus is that, for learned forelimb movements in the primate, motor cortex is likely to exert fairly direct control over muscles^{48–54,173}. Direct control is consistent with anatomy: M1 communicates with motor neurons more directly (as little as one synapse³²) than photoreceptors communicate with V1 (four synapses), and the spiking of an individual motor-cortex neuron can measurably impact muscle activity^{33,34}.

At first, there had been a sensible reason to doubt direct control of muscles: single-neuron responses do not resemble hand acceleration¹¹⁷, despite acceleration being proportional to force. Yet this turned out to be expected: given biomechanical complexities, muscle activity rarely mirrors hand acceleration during fast movements^{49,50,174}. Furthermore, given an expansive null-space, single-neuron activity is not expected to mirror muscle activity even when muscle activity is a primary readout (for example, Fig. 3c). Fortunately, studies could still test other predictions of direct control. Direct control correctly predicted that physical anisotropies, between reach direction and muscle-activity magnitude, would be reflected in motor cortex^{50,51}. It explained inconstancies in the lag between decoded and actual velocity⁴⁹, and successfully predicted that neural responses should be more closely tied to muscle-activity features than to abstract quantities such as hand direction^{48,54,91,175}. Direct control correctly predicted that, although neural ‘preferred directions’ will not mirror muscle preferred directions, they should exhibit related changes across postures⁵³. Finally, direct control successfully predicted that diverse individual-neuron activity should provide a good basis set for constructing muscle activity^{34,45,176,177}.

At the same time, there are reasons to presume that motor-cortex outputs are not identical to muscle activity⁴⁵. Descending drive may be altered within the spinal cord^{128,178} or brainstem¹⁷⁹. Furthermore, cortex provides only a portion of motor neuron drive, with subcortical

and spinal reflex circuits also contributing. Which aspects of muscle activity derive from which sources, and how this might vary across species and behaviours, remain open questions. Furthermore, in addition to driving motor neurons, descending cortical projections are likely to serve additional purposes, including modulating ascending sensory responses¹⁸⁰.

Is the null-space really null?

A core assumption of the factor-dynamics framework is the presence of null-space signals that are computationally important but do not encode movement parameters. This assumption distinguished the factor-dynamics framework from the traditional representational framework, where questions centred on which parameters are encoded in which reference frames^{181,182}. But perhaps there is no null-space. Single-neuron response complexity might instead reflect a plethora of encoded kinematic parameters (see ref. 183 for a review advancing this proposal). There are reasons to be sceptical of this proposition. Despite its longevity, the parametric-encoding hypothesis has yet to be instantiated in a concrete model that explains neural responses. Indeed, it has consistently been challenging for parametric-encoding models to generate responses that are realistic at both single-neuron and population levels. During cycling, standard parametric-encoding schemes are inconsistent with neural trajectories across directions³⁸ and speeds¹²³. During reaching, standard parametric-encoding models are inconsistent with single-neuron responses²² and factor-level rotations²³. Parametric-encoding models can be modified to create rotations (for example by introducing acceleration and velocity as represented parameters), but the resulting rotational frequencies are unrealistically high and the link between rotational phase and preparatory state still remains unexplained. Preparatory activity is not well explained by representational expectations⁶, nor are general properties of the population response. Population dimensionality opposes parametric-encoding predictions⁷⁶, and low trajectory tangling has no natural encoding-based explanation.

An advantage of factor-dynamics hypotheses is that they are typically explicit in modelling how neural activity might generate behaviour^{24,108,123}, facilitating behaviour-level predictions^{7,97,148,149}. In contrast, parametric-encoding models typically seek to describe responses, and are frequently vague regarding how represented parameters would be converted to muscle activity and thence behaviour. This is hardly a minor shortcoming, given that complex parametric-encoding schemes (such as that in ref. 184) would need to be decoded in real time⁴⁹ despite hypothesized encoding lags that are often acausal or inconstant (for example, ref. 185). Parametric encoding would also need to be task-specific; for example, correlations between neural activity and velocity are common during reaching¹⁸⁶ but not cycling¹⁸⁷.

Thus, we find factor-dynamics hypotheses, and their use of multiple output-null factors, to provide the most parsimonious explanations for single-neuron responses and population-level properties. An alternative is dynamics driven entirely by sensory feedback, with little or no need for a null-space during execution¹²¹. Models embodying this hypothesis produce realistic rotations of the neural state, and fairly realistic single-neuron responses (Fig. 4f). Such models do not explain why rotational dynamics precede sensory feedback, yet might be elaborated to do so by adding output-null factors. This illustrates that the field is close to moving past the debate regarding whether neural activity reflects parametric representations versus factor-level dynamics. Future inquiry will probably centre on the sources of factor-level

dynamics, the interplay between internal recurrence and sensory feedback^{95,101,188}, and the various roles of output-null factors.

Evaluating factor-level features

Moving forward, an increasingly common question will be whether a newly discovered factor-level feature (for example, trajectory rotations²³ or a putative trigger signal¹⁴⁸) is informative or would have occurred under nearly any reasonable hypothesis. Given uncertainty regarding the range of viable hypotheses, it can be difficult to establish expectations regarding which observations are expected versus surprising, especially for output-null factors. A common solution is surrogate-data controls, which resemble resampling methods but address whether a feature is surprising rather than whether it reflects sampling error. To create statistical stand-ins for alternative hypotheses, surrogate datasets are endowed with many properties of the data without directly building in the feature of interest. If the feature is stronger in empirical versus surrogate data, one hopes it may be 'surprising' in the sense of discriminating amongst hypotheses. For example, preparatory and execution subspaces are more distinct in empirical versus surrogate data, and factor-space rotations are stronger in empirical data than in various surrogate-data controls^{23,114}.

Surrogate-data controls do not reveal whether a feature is 'real' but, rather, whether it provides additional evidence for a hypothesis beyond what could (or should) have been inferred from other properties¹¹⁴. Consequently, surrogate-data controls involve subjectivity regarding which properties should be taken as a given¹¹⁴. Furthermore, surrogate-data controls cannot be counted on to capture what one really cares about: the distribution of possible results spanned by viable potential hypotheses. Thus, whenever possible, factor-level findings should be interpreted in the context of true competing hypotheses. For example, orthogonality of preparation and execution subspaces was striking because no existing model had predicted it. Similarly, factor-trajectory rotations were striking because they were not predicted by any prior hypothesis, but were predicted by a factor-dynamics model that also accounted for single-neuron responses²³. As hypotheses mature, it should become increasingly possible to evaluate them by directly comparing competing predictions, with 'surprise' relative to surrogate-data controls becoming a secondary consideration.

What should be in the null-space?

Motor cortex reflects some types of information while remaining invariant to others. For example, although the upstream supplementary motor area tracks progress towards a distant goal, motor cortex reflects only the present action¹⁸⁹. Similarly, motor-cortex activity is agnostic regarding whether a reaching movement is performed alone or begins a sequence⁸¹. Yet some contextual signals are present, including engagement¹⁹⁰ and reward expectation¹⁹¹.

There may be potential costs to having 'extra' output-null factors. Every factor reduces the dynamic range available for other factors, and learning may fail to generalize if the modified neural trajectory is context-dependent. Additionally, although output-null factors do not themselves impact readouts, they typically interact with factors that do. Such interactions may be difficult to eliminate even when undesirable. Large potential rewards appear to induce this phenomenon. When the reward-sensitive factor is at an extremum, the preparatory-factor state is distorted and behaviour suffers from paradoxically poor performance, reminiscent of 'choking under pressure'¹⁹¹. Why some contextual signals are present despite such potential costs remains a mystery.

Motor adaptation in the null-space

The goal of motor adaptation is to adjust outputs to compensate for a change in the body or environment. The factor-dynamics framework suggests multiple ways in which this could occur. Adaptation could alter the preparatory-factor state, factor-level dynamics linking preparation and execution, or final outgoing commands¹⁹². Given recurrent loops between motor cortex and the cerebellum⁹⁶, all three possibilities could involve cerebellar plasticity. Neural recordings indicate that the first possibility, and perhaps the second, does occur^{192–194}. Following a visuo-motor rotation, adaptation alters the mapping from visual inputs to the preparatory neural state, counteracting the rotation^{192,193}. On single trials, the degree of neural-state alteration predicts behavioural adaptation¹⁹³. Interestingly, not only does the preparatory neural state reflect adaptation, adaptation appears to depend upon the integrity of the preparatory state. Longer preparatory periods promote faster adaptation, and artificially disrupting the preparatory state impairs subsequent adaptation¹⁹⁵.

Adaptation to a curl-field evokes similar preparatory-state alterations, but also additional changes^{192,194}, presumably because curl-field forces cannot be fully counteracted by rotating the point where one aims. The additional changes are putatively output-null¹⁹², and involve a uniform shift of the entire pattern of preparatory states¹⁹⁴. Opposing curl-fields evoke uniform shifts in opposing directions. These shifts may relate to the phenomenon of separating skill variants using output-null factors¹⁹⁶ (Fig. 2d). By changing the overall location in factor-space, a shift could alter the local dynamics experienced by the factors¹⁹⁷.

Null-spaces and brain–computer interface decoding

Early brain–computer interface (BCI) decoding algorithms achieved success in reach-like tasks by leveraging directional signals that were hypothesized to be biologically output-potent^{198–201}. It currently seems unlikely that direction per se is the principal motor-cortex output. Yet incidental (and potentially useful) correlations with direction are expected under factor-dynamics hypotheses, and can be facilitated by output-null factors³⁸. This suggests that decoder-potent dimensions, when optimized to decode direction, are likely to overlap with the biological null-space. Null-space-based decoding may explain why subjects can cease moving yet still control the decoder (see ref. 199 for example). In some cases, decoding explicitly used null-space signals, including preparatory activity^{202–204} and output-null aspects of execution dynamics²⁰⁵.

The BCI field has had a long-standing interest in learning²⁰⁰, in part because experimenters can perturb the decoder, requiring subjects (typically monkeys) to learn to compensate. Because the decoder is known, one can determine with certainty which changes in neural activity alter the output. An embrace of null-spaces became central to this endeavour. Consider the linear decode: $\mathbf{c}(t) = \mathbf{B}^T \mathbf{r}(t) = \mathbf{B}^T \mathbf{U} \mathbf{x}(t)$, where $\mathbf{c}(t)$ is three-dimensional cursor velocity and $\mathbf{B}$ is an $N \times 3$ decoder matrix. Each column of $\mathbf{B}$ is a 'decoder-potent' neural dimension, containing velocity-decoding weights. Each column of $\mathbf{U}$ is a neural dimension describing factor tuning for one factor. If the experimenter alters $\mathbf{B}$, learning could compensate in two ways²⁰⁶. Factor tuning might adapt to compensate for the change in $\mathbf{B}$ (allowing $\mathbf{B}^T \mathbf{U}$ to remain constant). Alternatively, the subject could 'reassociate', altering the relationship between desired cursor movement and the factors, $\mathbf{x}(t)$. Pioneering studies perturbed the rows of $\mathbf{B}$ and found that learning was swift. Neural activity showed evidence of both learning strategies²⁰⁶ but reassociation was dominant²⁰⁷.

The null-space concept allowed refinement and extension of these results. If $\mathbf{x}(t)$ is higher-dimensional than $\mathbf{c}(t)$ (if $K > 3$), there exists a sizeable ($K - 3$ dimensional) ‘decoder-null’ space. If reassociation occurs in K -dimensional space, but tuning is computed in three-dimensional cursor-velocity space, tuning will appear to change even if $\mathbf{U}$ remains constant. Sadtler et al.¹⁶³ assumed a sizeable null-space: they estimated a ten-dimensional $\mathbf{x}(t)$ and employed a two-dimensional $\mathbf{c}(t)$. $\mathbf{B}^T$ was decomposed into $\mathbf{B}^T = \mathbf{QV}^T$, where $\mathbf{V}$ is an $N \times 10$ matrix of neural dimensions and $\mathbf{Q}$ is a 2×10 matrix mapping activity in those dimensions to screen coordinates. The perturbed decode was $\mathbf{B}^{\text{pert}T} = \mathbf{QP}\mathbf{V}^T$, where $\mathbf{P}$ is a 10×10 permutation matrix. Monkeys quickly compensated for most such perturbations, and learning was due entirely to reassociation^{163,208}. A key feature of reassociation is that it changes only how $\mathbf{x}(t)$ behaves, and not the factor tuning, $\mathbf{U}$, that determines the neural dimensions occupied by population activity. Thus, if monkeys rely primarily on reassociation, they should have difficulty compensating for arbitrary perturbations of the $N \times 2$ decoder matrix $\mathbf{B}$, because decoding dimensions would no longer align with any of the currently used factor-tuning dimensions. As predicted, learning to compensate for arbitrary perturbations took many sessions²⁰⁹, presumably reflecting slower reassociation involving previously unused factors, and/or synaptic plasticity²¹⁰ that alters $\mathbf{U}$.

In a complementary approach, Vyas et al.¹⁹³ used a BCI task that emulated a standard delayed-reach task, making it possible to determine the locus of BCI adaptation within the prepare–trigger–execute motif. Perturbations employed $\mathbf{B}^{\text{pert}T} = \mathbf{PB}^T$, where $\mathbf{P}$ is a 2×2 rotation matrix. Learning was swift, and relied on changes in the decoder-null preparatory state, which propagated into decoder-potent dimensions during movement. This strategy mirrored adaptation to visuo-motor rotations during normal reaching (see above). In agreement, learning transferred between BCI and non-BCI contexts. In another example of BCI adaptation involving a null-space²¹¹, monkeys were asked to adapt from decoder A to decoder B, then de-adapt back to decoder A. De-adaptation restored decoder performance, yet neural activity continued to show the influence of having learned decoder B. This result appears confusing from a single-neuron perspective. How can incomplete neural de-adaptation accompany complete behavioural de-adaptation? Yet this strategy is natural given a null-space: one can fully re-adapt to decoder A while retaining features appropriate for decoder B²¹¹.

The results above highlight an important engineering choice. Should BCI decoders intentionally leverage the null-space? This strategy has worked well in the past, but may impair generalization. For example, null-space signals that correlate with velocity may be prominent in some tasks but not others¹⁸⁷. An alternative strategy is to align decoder dimensions with the true output-potent neural dimensions. This could allow excellent generalization²¹², but faces the limitation that output-potent factors are likely to contribute a relatively small percentage of the total population variance^{38,187}. A final possibility is to abandon the idea of decoder-potent dimensions entirely, and use neural networks²¹³ or other highly non-linear means¹⁶⁹ to convert factor trajectories into decoded variables.

Unanswered questions regarding preparation

Motor preparation has long been proposed to take time, such that longer delays allow shorter reaction times^{14,137,138}. If so, the triggering input, arising from elsewhere in the brain, should presumably wait until the preparatory-factor state is accurate. In this view, preparatory factors are output-null only from the perspective of the muscles; preparatory

factors would be observed by other brain areas, perhaps allowing active refinement with time¹⁰⁹. These ideas successfully predicted that reaction times would be longer following atypical preparatory states⁹, and following perturbations of the preparatory state¹⁰.

It has since become clear that preparation is not necessarily time-consuming. For familiar movements, preparation can be extremely rapid^{81,135,144}. The trigger can occur so swiftly that it is unlikely to await preparatory refinement¹³⁵. It was thus argued that reaction times are completely independent of the preparatory-factor state^{144,214}. Does this mean that the trigger does not await preparatory refinement²¹⁵? Or is refinement simply negligible for familiar movements? One certainly suspects that unfamiliar or challenging movements would require preparatory refinement^{193,195}, but a general answer remains elusive. It is similarly unclear whether refinement (if present) involves loops between higher areas and motor cortex^{10,95,216}, or simply occurs upstream and is reflected in motor cortex^{145,215}.

An expansive null-space?

The null-space hypothesis received its first direct empirical test in the context of motor-cortex preparatory activity. Yet the null-space concept is both older and broader. Even before the key empirical test, studies linking network models to behaviour leveraged null-spaces implicitly (for example, refs. 22,154) and explicitly²⁹. Since that key test, the idea that some factors are output-potent (and thus communicate with other areas) while others are output-null (and thus do not) has been explanatory in the context of sensation (for example, refs. 31,217) and cognition (for example, ref. 218). Output-null factors are thought to serve multiple functions, even within motor cortex. In addition to preparation (Fig. 2a), potential functions include triggering of movement (Fig. 2b), reducing trajectory tangling (Fig. 2c), creating skill variants (Fig. 2d) and allowing the same network to instantiate multiple independent skills (Fig. 2e). In particular, the flexibility-via-subspace hypothesis depends on a null-space, and may be essential to understanding both motor flexibility and the cognitive flexibility underlying intelligence. The concept of a basis set, where diverse responses sum to produce a desired output, also assumes a null-space²¹⁹. As our understanding grows, it seems likely that the set of computational roles played by output-null factors will continue to expand. This would complete a sizable transition in the way that the field of motor control thinks about neural responses. Instead of viewing most aspects of neural responses as reflecting the encoding of movement parameters or outgoing commands, we will view most aspects of neural responses as reflecting output-null factors that do not directly drive outputs, but are nevertheless central to how the network does its job.

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Author contributions

The authors contributed equally to all aspects of the article.

Competing interests

M.M.C is an inventor of the MINT decoding approach, which has been licensed to Blackrock Neurotech. K.V.S. served on the Scientific Advisory Boards (SABs) of MIND-X Inc. (acquired by Blackrock Neurotech), Inscopix Inc. (merged with Brucker Nano) and Heal Inc.; served as a consultant/adviser for CTRL-Labs (on the founding SAB; acquired by Facebook Reality Labs in Fall 2019, which is now Meta Platform Reality Labs); and served as a consultant/adviser for and was a co-founder (2016–2023) of Neuralink.

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